When the Most Disruptive Messages Are Safe to Prune in Aggregate: Functional Analysis of Communication Pruning in Two-Agent LLM Systems

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Abstract

In two-agent LLM solver-checker systems, we uncover a *perturbation-sensitivity dissociation*: acknowledgments—the most individually disruptive message type—can be safely pruned in aggregate. We establish this finding across two backbone families (Qwen2.5-7B, Llama-3.1-8B), two reasoning domains, and multiple scales (7B primary, with preliminary 14B and 72B probes), confirmed by 10-seed replication ($p < 0.001$). Three acknowledgment detectors—an oracle classifier, a heuristic rule, and a learned DistilBERT model—all yield non-inferior pruned accuracy under non-inferiority testing, demonstrating that pruning safety is robust to detection method and precision.

Acknowledgment pruning eliminates 36–56% of inference calls (entire LLM invocations, not token-level compression) while maintaining task accuracy on GSM8K and StrategyQA. Correlational evidence is consistent with a *cascade dissolution* pattern in which self-reinforcing acknowledgment loops dissolve without degrading information flow. Boundary conditions—heuristic non-transferability, domain mismatch, and stochastic decoding sensitivity—motivate empirical screening before deployment. **All results are under greedy decoding in two-agent topologies.**

1 Introduction

LLM-based two-agent solver-checker systems communicate prolifically—a single math problem can generate dozens of messages—yet which are structurally necessary remains open. Existing approaches operate at different granularities: token-level compression Jiang et al. [2023], Chevalier et al. [2023], runtime trajectory filtering Lin et al. [2026], agent-level pruning Wang et al. [2025c], Zhang et al. [2025a], message-level suppression Li et al. [2025a], Zeng et al. [2025], and system-level optimization Chen et al. [2025]. Yet none perform *functional analysis*—understanding what communicative role each message plays—nor validate whether functional classification predicts which messages can be safely removed.

An intuitive hypothesis is common: high-information messages are important; low-information acknowledgments are expendable. This conflates *perturbation sensitivity*—how much removing a message disrupts flow—with *informational importance*—whether it carries task-relevant content. Pruning methods that score by content richness Wang et al. [2025c], Zeng et al. [2025] implicitly assume these correlate. Yet conversation analysis shows backchannels Yngve [1970], Schegloff [1982], Jefferson [1984] are structurally load-bearing despite negligible propositional content Clark and Schaefer [1989]. Whether LLM agents exhibit analogous dependencies has never been tested.

The gap. Three layers of evidence point in contradictory directions: (1) *information content* suggests acknowledgments are dispensable; (2) *individual perturbation* reveals they are structurally critical (91% high disruption); (3) *aggregate pruning* shows they are safely removable ($\Delta \approx 0$ pp). No prior work has tested this three-way paradox in LLM agents nor explained why individually critical messages become safely removable in aggregate Rizvi-Martel et al. [2026].

Contributions. To our knowledge, we present the first perturbation-based empirical study of message-level pruning in two-agent solver-checker LLM systems, spanning two primary backbones (Qwen, Llama; Gemma exploratory), two validated task domains (with ARC and MATH as supplementary), and three topologies (15 configurations at 7B, with 14B and preliminary 72B scale probes):

1. **Perturbation-sensitivity dissociation** (§5.1–5.3). Messages most disruptive to perturb individually can be pruned in aggregate without significant accuracy loss. We conjecture that self-reinforcing ack cascades create structural redundancy that dissolves under aggregate removal (Qwen 90.7%, Llama 61.9%)—a mechanism distinct from human backchannel dynamics, requiring causal verification. This finding is enabled by a reusable analysis protocol comprising a 5-class taxonomy, structured annotation, and perturbation-based impact measurement.
2. **Analysis protocol** (§4.2–4.4). A 5-class communicative function taxonomy validated by cross-model agreement (κ=0.721) and human annotation (κ=0.76), a structured annotation protocol (constrained JSON, cross-model validation), and a perturbation-based impact measurement pipeline with non-inferiority and TOST equivalence testing—designed to be applicable to any two-agent LLM system for message-level communication analysis. The taxonomy serves a dual role: as a scientific lens for understanding communication structure, and as a label generator enabling automated pruning without per-message annotation at deployment.
3. **Pruning: from oracle to deployment** (§5.2, 5.5). Oracle and annotation-free heuristic pruning reduce inference calls by 36–56% with non-inferiority at $\delta = -3$ pp ($p < 0.001$) and supplementary Qwen-specific TOST equivalence at ± 1 pp (oracle $p = 0.048$, heuristic $p = 0.007$; 10 seeds each), validated across two backbone families and two task domains under greedy decoding. Boundary conditions—heuristic non-transferability (Qwen3), domain mismatch (MATH), and backbone-dependent stochastic decoding sensitivity—motivate empirical screening ($N \geq 200$) and a learned-classifier path beyond hand-crafted rules.

Scope. Claims are validated under greedy decoding; stochastic decoding (t=0.7) is validated for Qwen (10 seeds, TOST $p < 0.001$ at $\delta = -3$ pp) with Llama showing a marginal degradation trend (§5, §6). Our message-level analysis complements token-level compression Jiang et al. [2023] and agent-level pruning Wang et al. [2025c], bridging theoretical bounds Rizvi-Martel et al. [2026] and practical pruning. The taxonomy serves as an analytical lens; heuristic detectors that make it unnecessary at deployment are products of the analysis (§2).

2 Related Work

Our work draws on conversation analysis, multi-agent communication, and message function analysis.

Conversation analysis and grounding. Human conversation relies on low-content but structurally essential signals that coordinate mutual understanding. Backchannels Yngve [1970], continuers Schegloff [1982], and acknowledgment tokens Jefferson [1984] carry little propositional content yet are load-bearing in conversational grounding Clark and Schaefer [1989]; Bavelas *et al.* Bavelas et al. [2000] demonstrate that reducing listener backchannels degrades narrative quality, confirming their functional necessity. Computational extensions have addressed backchannel prediction Liermann et al. [2023], Inoue et al. [2025] and representation learning Qian and Skantze [2026], but exclusively in human conversation; whether analogous functional dynamics hold for LLM-generated acknowledgments remains untested.

Multi-agent communication. Research on multi-agent communication spans from structured protocols to emergent language, addressing different facets of *how*, *when*, and *what* agents should exchange. Early work defined formal message types through agent communication languages Finin et al. [1994], while learned communication Sukhbaatar et al. [2016], Foerster et al. [2016], Wang et al. [2020] optimized bandwidth end-to-end. Recent theoretical contributions address complementary dimensions: Rizvi-Martel *et al.* Rizvi-Martel et al. [2026] derive bounds on *when* communication helps, Zheng *et al.* Zheng et al. [2025] explore *alternatives* such as latent thoughts, and Chen *et* *al.* Chen et al. [2026b] identify "what to communicate" as a key open question. At the application level, multi-agent debate Du et al. [2024] and LLM orchestration frameworks such as CAMEL Li et al. [2023], MetaGPT Hong et al. [2024], AutoGen Wu et al. [2024], AgentVerse Chen et al. [2024], and MegaAgent Wang et al. [2025a] confirm that communication structure affects downstream performance, though debate may collapse to majority voting Choi et al. [2025] and intrinsic model strength can dominate architectural choices Wu et al. [2025], Kaushal and Singh [2026], La Malfa et al. [2025]; see Yan *et al.* Yan et al. [2025] and Tran *et al.* Tran et al. [2025] for surveys. Zhou *et al.* Zhou et al. [2025b] and Buscemi *et al.* Buscemi et al. [2025] investigate *why* agents communicate but neither performs message-level functional analysis. Our work fills this gap by studying which natural-language messages are safe to *remove* based on their communicative function.

Communication efficiency. Existing efficiency methods operate at different granularities, but none analyze the communicative *function* of individual messages. At the coarsest level, Agent-Dropout Wang et al. [2025c, 2026] prunes entire agents, achieving 20–56% token reduction without distinguishing which *messages* are dispensable. Finer-grained approaches target turns and edges: S^2-MAD Zeng et al. [2025] applies turn-level early stopping, OPTIMA Chen et al. [2025] optimizes topology and routing, and SafeSieve Zhang et al. [2025c] prunes inter-agent edges via dual-mechanism scoring. Lin *et al.* Lin et al. [2026] introduce SupervisorAgent, a lightweight meta-agent that uses LLM-free adaptive filtering at runtime to reduce token consumption by $\sim 30\%$, though their filtering operates at the token/trajectory level rather than at the level of communicative function. Complementary directions include agent pruning Zhang et al. [2025a], Li et al. [2025a], Xiao et al. [2026], Wang and Tong [2025], latent-space communication Zou et al. [2025], Du et al. [2025b], KV-cache sharing Ye et al. [2025], and holistic optimization Li et al. [2025b], Zhang et al. [2025b], Jiang et al. [2026] (Appendix Table 34). In contrast, our taxonomy-driven pruning targets specific message categories identified through functional analysis, and we provide the first evidence that pruning safety is invariant across a $10\times$ parameter range (7B–72B).

Topology and protocol. A parallel thread investigates *who* communicates and *how*, optimizing organizational structure rather than message content. Automated topology design methods such as GoAgent Chen et al. [2026a] and AnyMAC Wang et al. [2025b] search over communication graphs, while LACP Li et al. [2025c] and GUARDIAN Zhou et al. [2025a] learn adaptive protocols. Shen *et al.* Shen et al. [2025] analyze how topology shapes information propagation. Benchmarks including ProtocolBench Du et al. [2025a] and MultiAgentBench Zhu et al. [2025] provide standardized evaluation for these structural choices. We complement this body of work by analyzing *what* is communicated at the message-function level, an orthogonal dimension that topology-focused methods leave unexamined.

Attribution and pragmatics. Attribution methods for multi-agent systems—including Hive-Mind Xia et al. [2026], AgentSHAP Horovicz [2025], and MAST Cemri et al. [2025]—assign credit at the agent or trace level, but do not decompose individual messages by communicative function; our ablation operates at this finer granularity. Our functional taxonomy draws on classical speech act theory Austin [1962], Searle [1969] and established dialogue act annotation schemes such as DAMSL Core and Allen [1997] and ISO 24617-2 Bunt et al. [2012]. Recent work has begun applying these frameworks to LLM settings: Qamar *et al.* Qamar et al. [2025] annotate dialogue acts in LLM conversations, Maitra *et al.* Maitra et al. [2025] use them as a lens on human–LLM interaction quality, and Zhang *et al.* Zhang et al. [2025b] find that acknowledgment and hints play key roles in dual-agent math problem-solving. We extend this line of inquiry to inter-agent communication, connecting functional categories directly to pruning outcomes.

3 Taxonomy of Communication Functions

We define five exhaustive, mutually exclusive communication function types (Table 1). Ambiguity is resolved by hierarchical priority: *coordination* $\succ$ *verification* $\succ$ *reasoning_guidance* $\succ$ *information_provision* $\succ$ *acknowledgment*, reflecting the principle that structurally higher-level functions take precedence.

Our categories map onto established dialogue act frameworks (DAMSL Core and Allen [1997], ISO 24617-2 Bunt et al. [2012], FIPA ACL Foundation for Intelligent Physical Agents [2002]; detailed mapping in Appendix N), collapsing 56+ fine-grained types into five categories operationalized via

Table 1: Five-class taxonomy of communication functions in solver–checker dialogues, with definitions and GSM8K examples. Priority: coordination $\succ$ verification $\succ$ reasoning $\succ$ information $\succ$ acknowledgment.

Function Type	**Definition**	**Example**
INFORMATION PROVISION	Supplies factual content, data, or intermediate results	"The total cost is 3 × \$5 = \$15 for the shirts."
REASONING GUIDANCE	Directs reasoning through decomposition or strategic suggestions	"First calculate the weekly cost, then multiply by 4."
VERIFICATION	Checks or corrects the correctness of prior messages	"17 × 3 = 51, not 53. The corrected answer is \$51."
COORDINATION	Manages dialogue flow, assigns roles, or delegates tasks	"I'll solve first, then you verify step by step."
ACKNOWLEDGMENT	Simple confirmations or agreements with no new content	"Your solution is correct."

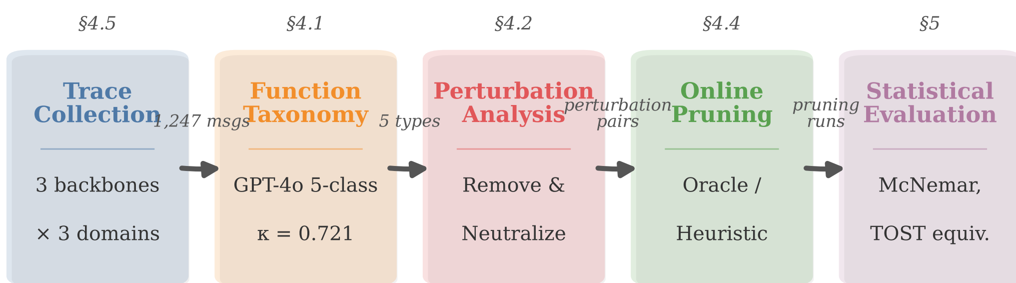

Figure 1: Experimental pipeline. Multi-agent conversations (2+1 backbones $\times$ 3 domains, 15 configurations) are annotated into a 5-class taxonomy (cross-model κ=0.721), subjected to single-message perturbation analysis (remove and neutralize ablation), then pruned online under oracle or heuristic detection, and evaluated via McNemar exact tests and TOST equivalence.

measurable perturbation sensitivity rather than speaker intent. The taxonomy was developed for two-agent solver-checker dialogue; applicability to other configurations remains untested.

Each message is annotated by an LLM annotator using a structured prompt with taxonomy definitions and hierarchical decision rules (temperature 0, constrained JSON; Appendix K). Cross-model validation uses two independent annotator models per message ($\kappa \geq 0.4$ threshold; Section 5.4).

4 Experimental Setup

4.1 Analysis Protocol Overview

Existing MAS communication analyses operate at the turn level (which agent speaks when) or the system level (total token count). Neither captures the functional heterogeneity *within* turns: a single checker response may acknowledge, verify, and redirect in successive sentences. We therefore adopt message-level granularity, classifying each message by its communicative function rather than its surface form.

The 5-class taxonomy—acknowledgment, verification, reasoning guidance, information provision, coordination—is derived from speech act theory Searle [1969] and conversation analysis Schegloff [1982], Jefferson [1984], adapted to the structured interactions of LLM solver-checker systems. Five classes balance expressiveness against annotator reliability: coarser schemes (ack vs. non-ack) lose the impact gradient; finer schemes degrade inter-annotator agreement below actionable thresholds. The analysis protocol is designed for reuse: researchers can apply the taxonomy schema, annotation protocol, and statistical evaluation pipeline to any LLM-based multi-agent system without redesigning the classification or assessment methodology.

4.2 Annotation Protocol

The annotation protocol is designed for reproducibility and portability across MAS configurations. Using the taxonomy from Section 3, each message is annotated via structured prompt with taxonomy definitions and hierarchical decision rules (temperature 0, constrained JSON output). To avoid self-annotation bias, messages generated by backbone A are annotated by backbone B. Non-blinding could inflate ack classifications but makes the pruning test more conservative. The protocol requires only a taxonomy-aware LLM annotator and constrained decoding—no task-specific training data—making it directly applicable to new backbone or domain configurations. Our core claims depend on the ack/non-ack binary distinction (Cohen's κ=0.721 Cohen [1960], substantial agreement); fine-grained per-type agreement varies ($\kappa > 0.8$ for acknowledgment, κ=0.237 for information_provision), so sub-category analyses are labeled preliminary. A human validation study ($N = 225$, $\kappa = 0.76$) is reported in Section 6.

4.3 Perturbation-Based Impact Measurement

To isolate each message's contribution to downstream behavior, we introduce two complementary ablation procedures applied independently to every message in a recorded conversation trace.

Remove ablation. Message m_i is deleted from trace $\mathcal{T}$, and all subsequent messages are discarded and regenerated from position $i + 1$ onward. The behavioral impact is:

$$\text{sim}_{\text{remove}}(m_i) = \cos\big(\mathbf{e}(a_{\text{orig}}),\, \mathbf{e}(a_{\text{regen}})\big) \tag{1}$$

where a_{orig} denotes the final solver answer in the original trace and a_{regen} the answer regenerated from position i+1 after removing m_i; $\mathbf{e}(\cdot)$ denotes the sentence embedding from `all-MiniLM-L6-v2` Reimers and Gurevych [2019]. Cosine similarity measures perturbation sensitivity—behavioral disruption from message removal. This is distinct from pruning safety, which is independently evaluated via task accuracy and McNemar's exact test (§4.5, Table 2). The impact gradient (§5.1) characterizes perturbation patterns; it does not serve as the basis for pruning safety claims.

Neutralize ablation. Message m_i is replaced with a type-matched placeholder that preserves structural position while removing informational content. Comparing remove and neutralize ablation disentangles a message's *content* from its *structural presence*.

Each ablation is a single-message perturbation using greedy decoding via vLLM Kwon et al. [2023] on 4× NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each), with conversations limited to $K = 6$ rounds.

Operational definitions. *Perturbation sensitivity*: $1 - \text{sim}_{\text{remove}}(m_i)$ (Eq. 1). *Informational importance*: task-relevant content operationalized via bigram novelty, token length, error correction rate, and Jaccard overlap.

4.4 Statistical Framework

The evaluation pipeline generalizes beyond acknowledgment pruning: any message-level intervention in a multi-agent system can be assessed by (1) measuring per-type perturbation sensitivity via single-message ablation, (2) testing aggregate pruning safety via McNemar's exact test and non-inferiority bounds, and (3) quantifying equivalence via TOST. We instantiate this pipeline as follows.

We use OLS ANCOVA (Equation 2) with function type as the factor and position/round as covariates, plus linear mixed-effects (ME) models (Equation 3) with task as a random intercept (the maximal converging specification):

$$y_i = \beta_0 + \textstyle\sum_k \beta_k \cdot \text{FT}_{ik} + \gamma_1 \cdot \text{Pos}_i + \gamma_2 \cdot \text{Round}_i + \epsilon_i \tag{2}$$

$$y_{ij} = \beta_0 + \sum_k \beta_k \cdot \mathrm{FT}_{ijk} + \gamma_1 \cdot \mathrm{Pos}_{ij} + \gamma_2 \cdot \mathrm{Round}_{ij} + u_j + \epsilon_{ij} \tag{3}$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$. Per-configuration ME models use individual tasks as random intercepts (maximal converging specification). Function type significance is tested via LRT with Benjamini–Hochberg correction ($\alpha = 0.05$; nine primary tests). End-to-end pruning safety is assessed primarily via confidence intervals on accuracy deltas (e.g., Qwen oracle 10-seed 95% CI $[-0.33, +1.13]$ pp) and McNemar's exact test McNemar [1947]. Non-inferiority testing at $\delta = -3$ pp provides supplementary evidence: at baseline $\sim$90%, this margin represents a $\sim$30% relative increase in error rate (10% $\rightarrow$ 13%), calibrated against observed seed-to-seed variance (SD $\approx$ 1–2.4 pp) to separate genuine degradation from sampling noise (Appendix S.2) Lakens [2017]. TOST equivalence at ± 1 pp further quantifies precision where applicable (Qwen oracle $p = 0.048$, marginal; interpret alongside the CI).

Multiple comparison framework. We organize statistical tests into three families to control false discovery risk while preserving power. *Family A* (primary confirmatory): nine LRT tests for function-type effects in mixed-effects models receive Benjamini–Hochberg correction at $\alpha = 0.05$. *Family B* (online cross-domain): McNemar tests across the backbone$\times$domain pruning matrix receive BH correction within family. *Family C* (offline exploratory): single-run offline comparisons (e.g., FG vs. ack-only, $N{=}100$) are explicitly labeled exploratory and do not receive cross-family correction. This hierarchy follows the confirmatory/exploratory distinction recommended in pre-registration practice; all Family C p-values are reported for transparency but should not be interpreted as confirmatory evidence.

4.5 End-to-End Pruning Protocol

We use *pruning* to refer to three distinct operations: (1) *remove-one ablation*—removing a single message to measure local impact (§4.3); (2) *offline replay*—removing all messages of a given type and replaying the conversation; (3) *online suppression*—intercepting and withholding messages in real time during live agent interaction. Context disambiguates throughout. Messages are suppressed per strategy and the conversation suffix is regenerated from the first suppression point. Four strategies are compared (§5.2): Baseline, Ack-only, FG (lowest-impact function type), and Random (10 seeds). Each "seed" controls problem selection (greedy decoding is deterministic).

Oracle detection. Two oracle pipelines establish upper bounds with perfect function-type detection. The *offline* pipeline precomputes GPT-4o labels on baseline conversations; the *online* pipeline annotates each message in real time within the pruned trajectory, eliminating distribution-shift confounds at higher cost. GPT-4o vs. GPT-4o-mini validation ($\kappa = 0.822$; Appendix S.21) confirms a cost-effective substitute.

4.6 Heuristic Detector Development

Because backbone choice determines communication surface form (§5.4), we develop *model-adapted* heuristic detectors that require no LLM annotation—analogous to task-specific prompt engineering, where rules are tuned to the target system rather than designed for universal transfer.

Qwen detector. A character-length threshold (< 30) exploits Qwen's narrow ack surface band (42.2% of checker messages below 30 chars, median 4 tokens; Appendix T), yielding F1 $= 0.845$ on GSM8K with $\sim$42% suppression.

Llama detector. Llama's verbose acks (mean 59.8 tokens, 65.6% bigram novelty; Appendix T) render length thresholds near-vacuous. A composite detector combines keyword matching, 4-gram repetition ratio (> 0.3), and a relaxed length ceiling (< 300 chars), yielding $\sim$75% suppression with comparable pruning safety (§5.2).

Both detectors are task-specific proof-of-concept classifiers; limitations are evaluated in Section 5.5 and discussed in Section 6.

4.7 Learned Classification

To quantify the gap between hand-crafted heuristics and learned detectors, we train supervised classifiers on the binary ack/non-ack label derived from the 5-class oracle annotations. The training corpus comprises 6,069 messages spanning all three backbones (Llama, Qwen, Gemma) and four domains (GSM8K, StrategyQA, ARC, MATH). We split by `task_id` hash (70/20/10 train/val/test) to prevent messages from the same problem instance from appearing in both training and evaluation sets—a critical design choice, since messages within a task share surface cues that would inflate held-out scores under random splitting.

TF-IDF + logistic regression. Character n-grams (n=2–5, `char_wb`) and word n-grams (n=1–2) are extracted via TF-IDF and fed to a balanced logistic regression classifier. This lightweight pipeline achieves test-set F1 $= 0.859$ (macro-averaged, N=526), substantially outperforming the best heuristic detector (F1 $= 0.542$), confirming that learned features capture ack surface patterns that hand-crafted rules miss across backbone–domain combinations.

Fine-tuning DistilBERT Sanh et al. [2019] on the same split achieves test-set F1 $= 0.867$ (macro; ack-class 0.822), comparable to TF-IDF+LR, confirming the lightweight pipeline as a practical alternative.

The cross-backbone, cross-domain training set ensures that classifiers generalize beyond any single configuration. Importantly, oracle pruning safety (§5.2) does not depend on detector quality—it establishes an upper bound under perfect detection—so learned classifiers serve as practical deployment tools rather than prerequisites for the pruning result.

4.8 Cross-Domain Experimental Design

A 3×3 factorial design (two primary backbones + one exploratory $\times$ three domains: GSM8K Cobbe et al. [2021], ARC-Challenge Clark et al. [2018], StrategyQA Geva et al. [2021]) yields 15 configurations (GSM8K under RR/SEQ/SEL; ARC and StratQA under RR only).

4.9 Implementation and Evaluation

Models and infrastructure. Two primary backbones—Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct—are selected for their contrasting communication styles: terse acknowledgments (Qwen median 4 tokens) versus verbose ones (Llama mean 59.8 tokens; §5.1). Scale probes at 14B (Qwen2.5-14B) and 72B (Qwen2.5-72B) test generalization beyond the 7B regime; Gemma-2-9B-it serves as an exploratory boundary condition for heuristic transferability (§5.5). Each agent team comprises a solver and a checker served via vLLM v0.6.6 Kwon et al. [2023] with greedy decoding (temperature $= 0$, top-$p = 1.0$) on 4× NVIDIA RTX PRO 6000 GPUs (96 GB each), orchestrated through AG2 v0.6.1 `ConversableAgent` with task-specific system prompts and a 10-round limit. Under selector topology, a third agent routes turns via `GroupChat` with automatic speaker selection. Function-type labels are assigned by two independent LLM annotators (Qwen2.5-7B as Model A, Llama-3.1-8B as Model B) with structured JSON output; oracle ablation and cross-model validation use GPT-4o (`gpt-4o-2024-08-06`). The 3×3 factorial design (§4.8) yields 15 configurations; combined with oracle, heuristic, and learned pruning strategies across multiple seed counts, the study spans approximately 120 GPU-hours.

Datasets and correctness criterion. Primary evaluation uses GSM8K Cobbe et al. [2021] (grade-school math, N=200–1,000 per configuration) and StrategyQA Geva et al. [2021] (commonsense reasoning, N=200–238). ARC-Challenge Clark et al. [2018] (N=250) serves as a cross-domain probe; MATH Hendrycks et al. [2021] (N=200, 102 valid after filtering) provides a boundary-condition test. Solution correctness is evaluated by exact match: extracted numerical answer for GSM8K, yes/no for StrategyQA, and letter for ARC-Challenge.

Evaluation metrics. Pruning effectiveness is quantified along two axes:

- *Accuracy delta*: $\Delta = \mathrm{acc}_{\text{pruned}} - \mathrm{acc}_{\text{baseline}}$, in percentage points (pp). Pruning is deemed *safe* under the non-inferiority criterion $\Delta \geq -3$ pp (§4.4).
- *Suppress rate*: $r_{\text{sup}} = n_{\text{suppressed}} / n_{\text{total}}$, the fraction of checker messages detected as acknowledgment and withheld during online pruning.

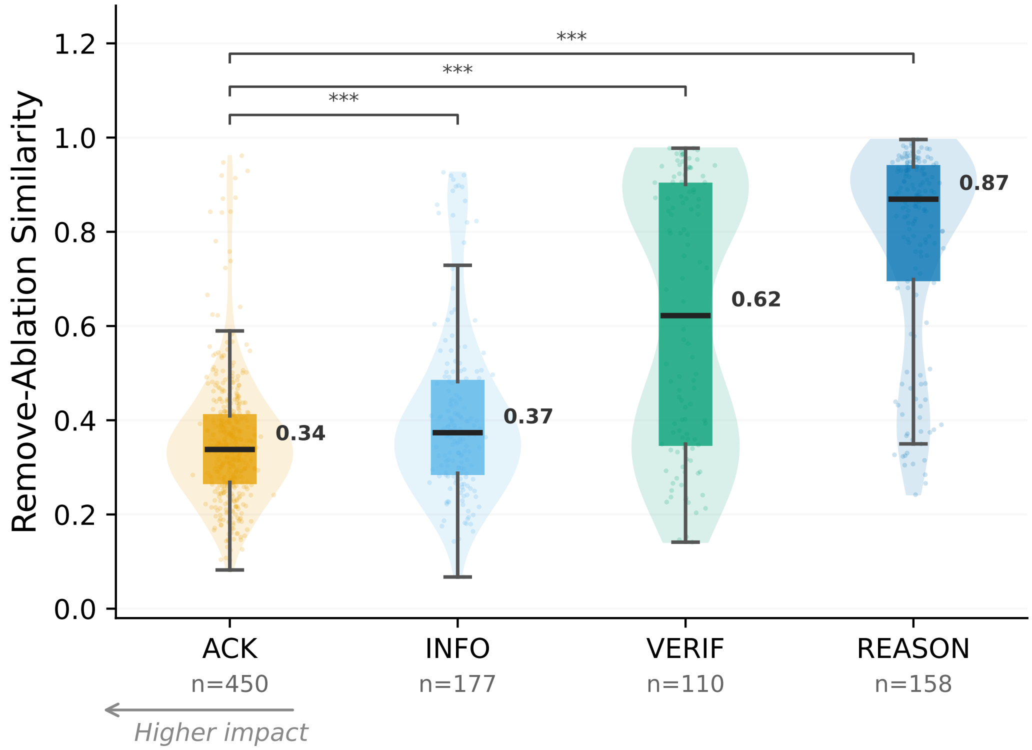

Figure 2: Remove-ablation similarity by function type (Qwen2.5-7B, round-robin, $n = 895$). Lower = greater disruption. Acknowledgments (median $= 0.34$) are most impactful despite least information content.

For single-seed configurations, statistical significance is assessed via McNemar's exact test McNemar [1947] on the 2×2 concordance table of per-problem correct/incorrect outcomes between baseline and pruned runs (§4.4). Multi-seed configurations additionally report paired t-tests and 95% confidence intervals on seed-level Δ values, enabling TOST equivalence testing Lakens [2017].

Replication strategy. Each "seed" controls problem sampling order; generation is deterministic under greedy decoding. The flagship configuration (Qwen oracle, GSM8K) uses 10-seed replication ($10 \times 200 = 2{,}000$ task instances) to support tight confidence intervals and TOST equivalence testing. Secondary configurations (Llama GSM8K, both backbones on StrategyQA) use 5–8 seeds. Scale probes (72B) and boundary-condition tests (MATH, temperature $= 0.7$) are single- or few-seed and explicitly labeled exploratory. Code and data are available in the anonymous repository at `[anonymized]`.

5 Results

We report results across 15 two-agent backbone×domain×topology configurations. The central question: in two-agent solver-checker systems, does perturbation sensitivity predict pruning safety?

5.1 Impact Gradient

For Qwen under fixed-turn topologies (Figure 2), the impact gradient is inversely related to information content: ACKNOWLEDGMENT (mean similarity $= 0.352$, $n = 450$), INFORMATION_PROVISION (0.410, $n = 177$), VERIFICATION (0.616, $n = 110$), REASONING_GUIDANCE (0.778, $n = 158$). Informationally sparse acknowledgments are those whose removal is most disruptive.

Pruning evidence. Removing a single acknowledgment is associated with high disruption (similarity < 0.5) in 91% of cases vs. 19% for reasoning guidance. The neutralize ablation produces the *opposite* gradient: neutralized information provision is associated with maximal disruption (0.047) while neutralized acknowledgments are associated with only moderate disruption (0.153; Appendix G). This double dissociation confirms acknowledgment impact stems from structural presence, not content. Robustness: placeholder choice $\rho = 0.73$; non-ack ρ=0.90; four similarity metrics $\rho > 0.91$ (Appendix I); RR/SEQ consistent, selector breaks down.

Taxonomy-independent information measures. Independent of taxonomic labels, acknowledgments are quantifiably information-sparse: Qwen acks have 5.7% bigram novelty (vs. 74.9% for reasoning guidance), median 4 tokens (vs. 117), and Jaccard overlap 0.69. Llama acks are longer (mean 59.8 tokens, 65.6% bigram novelty) but exhibit 0% error correction yet 14.3% error injection—a one-directional error channel (Appendix T).

5.2 E2E Communication Pruning

The individual perturbation results above suggest acknowledgments should be critical to preserve. The aggregate pruning tests below reveal the opposite—a dissociation we analyze in Section 6. We compare four strategies: **Baseline**, **Ack-only** (remove acknowledgments), **Function-Guided (FG)** (remove lowest-impact function type), and **Random** (same count as ack-only, 10 seeds).

Main results. Despite their individual disruptive impact, acknowledgments can be safely pruned in aggregate (Table 2, Figure 3). Both backbones show non-inferiority at $\delta = -3$ pp ($p < 0.001$). Ten-seed Qwen oracle: $\Delta = +0.40 \pm 1.02$ pp (95% CI $[-0.33, +1.13]$ pp; TOST $p = 0.048$ at ± 1 pp; 56.0% suppression). Ten-seed Llama oracle: $\Delta = +2.1 \pm 2.4$ pp (95% CI $[+0.4, +3.8]$ pp; 9/10 positive). The positive direction reflects a one-directional error channel (0/1,483 corrections but non-zero injection; Fisher $p = 0.004$). Non-inferiority ($p < 0.001$) provides the primary safety evidence; TOST provides supplementary equivalence precision (Qwen oracle $p = 0.048$, marginal; interpret alongside CI).

Message-level suppression (36–56%) translates to lower token-level savings ($\sim$10% Qwen, $\sim$28% Llama; Appendix W) because acknowledgments are shorter than substantive messages; the primary efficiency gain is inference-call elimination, not token reduction.

Heuristic detectors confirm deployment feasibility: Qwen 10 runs $\Delta = -0.05 \pm 0.99$ pp (TOST $p = 0.007$ at ± 1 pp). Llama 5 seeds $\Delta = +2.20 \pm 1.44$ pp (95% CI $[+0.41, +3.99]$ pp; non-inferiority $p < 0.001$): pruning is not merely non-inferior but suggestive of a small beneficial effect, though the interval width and limited seed count (5) preclude a strong improvement claim. The heuristic over-suppresses ($\sim$75% vs. $\sim$37% oracle rate), yet the positive direction suggests over-suppressed messages are predominantly redundant. Qwen StrategyQA (5 seeds) yields $\Delta = +0.70 \pm 0.91$ pp (95% CI $[-0.43, +1.83]$), consistent with non-inferiority (p < 0.001 at $\delta = -3$ pp). The Llama StrategyQA heuristic (8 seeds) yields $\Delta = +2.81 \pm 3.94$ pp (95% CI $[-0.49, +6.11]$), directionally positive but with high seed-level variance; non-inferiority holds (CI lower bound > -3 pp) with suppress rate stable across seeds (49–52%). Under stochastic decoding (t=0.7), pruning safety is backbone-dependent: Qwen oracle ($\Delta = +0.30 \pm 1.80$ pp, 10 seeds; TOST $p < 0.001$ at $\delta = -3$ pp) remains safe, while Llama shows marginal degradation (pooled $\Delta = -2.2$ pp, Fisher $p = 0.29$, NS; Appendix C).

In exploratory offline comparison (N=100, Qwen), FG pruning showed a substantial accuracy gap relative to ack-only (-5.0 pp vs. $+1.0$ pp, $p = 0.031$, exploratory), indicating aggregate safety is ack-specific. Online FG is confounded by cascading suppression ($\sim$90% removal). Robust to 20% label noise (Appendix S).

Token-matched random baseline. Online random (N=500, Qwen) matches ack-only ($\Delta = +0.8$ pp, NS), but ack-only suppresses $5.7\times$ more messages (56.3% vs. 9.9%) without per-task calibration. Per-task paired comparison confirms near-identical outcomes: only 4/500 tasks differ between strategies (McNemar $p = 1.0$; ack risk of correct$\rightarrow$wrong flip: 0.7%, identical to random),

[1] †TOST ± 1 pp (marginal). ‡Non-inferiority $\delta = -3$ pp. §Context overflow. ¶N=1,939 after filtering ($\sim$3% loss). ⋆Exploratory; confounded: suppress rate (79.3%) $\gg$ oracle ack rate (<5%); result reflects message redundancy, not ack-specific pruning.

Table 2: Online suppression pruning accuracy (%, N tasks, McNemar exact p). All rows use online suppression (messages intercepted during live agent interaction; cf. remove-one ablation in §4.3, offline replay in Appendix S). *Oracle*: GPT-4o 5-class ack-only. *Heuristic*: annotation-free rules (F1 $= 0.845/0.805$). Multi-seed rows report $\Delta \pm$ SD (pp). Supp. = messages suppressed.[1]

Setting	N	**Baseline**	**Pruned**	p	**Supp.**
Oracle (GPT-4o 5-class, ack-only)					
Llama / GSM8K	500	82.2	81.6 (−0.6)	.710	36.6%
Qwen / GSM8K	500	88.8	89.6 (+0.8)	.340	56.3%
Llama / GSM8K (t=0.7)	500	82.2	79.2 (−3.0)	.150	28.0%
Qwen / GSM8K (t=0.7)	500	87.4	88.0 (+0.6)	.710	53.1%
Qwen / GSM8K (t=0.7, 10 seeds)	10×200	$\Delta = +0.30 \pm 1.80$ pp		<.001‡	39.2%
Qwen / StratQA	500	73.2	73.4 (+0.2)	1.000	65.6%
Llama / StratQA§	238	68.9	73.9 (+5.0)	.180	21.5%
Qwen / ARC	250	90.8	91.6 (+0.8)	.500	14.6%
Qwen / GSM8K (10 seeds)	10×200	$\Delta = +0.40 \pm 1.02$ pp		.048†	56.0%
Llama / GSM8K (10 seeds)	10×200¶	$\Delta = +2.1 \pm 2.4$ pp		<.001‡	36.5%
Token-matched random (online control)					
Qwen / GSM8K	500	88.8	89.6 (+0.8)	.340	9.9%
Heuristic (annotation-free, exploratory)					
Llama / GSM8K (5 seeds)	5×200	$\Delta = +2.20 \pm 1.44$ pp		<.001‡	74.9%
Qwen / GSM8K (10 runs)	10×200	$\Delta = -0.05 \pm 0.99$ pp		.880	41.8%
Qwen / StratQA (5 seeds)	5×200	$\Delta = +0.70 \pm 0.91$ pp		<.001‡	56.9%
Llama / StratQA (8 seeds)	8×200	$\Delta = +2.81 \pm 3.94$ pp		.002‡	50.3%
Qwen / GSM8K (t=0.7)	500	88.6	88.0 (−0.6)	.730	37.3%
Llama / GSM8K (t=0.7)	500	81.2	77.8 (−3.4)	.100	59.3%
Gemma / GSM8K⋆	300	75.3	76.0 (+0.7)	.683	79.3%
Length-only baseline (char < 30)					
Llama / GSM8K (5 seeds)	5×200	$\Delta = -0.10 \pm 1.29$ pp		—	0.8%
Qwen / GSM8K (3 seeds)	3×200	$\Delta = +0.50 \pm 1.50$ pp		—	42.2%
Learned classifier (DistilBERT, F1 = 0.867)					
Qwen / GSM8K	300	89.3	90.0 (+0.7)	.727	44.8%

demonstrating that acknowledgments are genuinely information-free—removing $5.7\times$ more of them produces zero additional harm. For Llama, ack-only significantly outperforms matched-count random removal (83% vs. 73.2%), ruling out conversation shortening as the explanation.

Length-only baseline. A character-length filter (<30 chars) yields backbone-divergent results (Table 2). For Qwen, it matches composite heuristic performance ($\Delta = +0.50 \pm 1.50$ pp, 3 seeds; $p = 0.59$), confirming length as the primary discriminative axis. For Llama, the filter is near-vacuous (0.8% suppression) while the composite heuristic suppresses $\sim$75%, confirming that keyword and repetition rules are essential (§5.3).

Learned classifier online validation. Replacing heuristic rules with a fine-tuned DistilBERT classifier (§4.7; F1 $= 0.867$) in the online pruning loop yields $\Delta = +0.7$ pp (89.3% $\rightarrow$ 90.0%, N=300, McNemar $p = .727$, NS; 44.8% suppression; Table 2). Under our non-inferiority framing, this non-significant result is the expected outcome: the McNemar test detects no accuracy difference between pruned and baseline conditions, consistent with the NS pattern across all single-seed oracle and heuristic configurations (Table 2). The discordant pair profile ($b = 3$, $c = 5$) reinforces this interpretation: correct-to-wrong flips are rare (1.0%) and nearly balanced by wrong-to-correct gains, indicating no directional harm from classifier-guided pruning. That pruning safety holds across three detector types—heuristic rules (F1 $= 0.805$–0.845), learned classifier (F1 $= 0.867$), and GPT-4o oracle—suggests safety depends on the pruning *target* (acknowledgments) rather than detector precision. Compared to the Qwen heuristic ($\Delta = -0.05$ pp, 41.8% suppression), the learned classifier achieves slightly higher suppression (44.8% vs. 41.8%) while maintaining equivalent accuracy, demonstrating a practical drop-in replacement for annotation-free rules. DistilBERT inference adds $\sim$10 ms per message on CPU, negligible relative to vLLM generation latency.

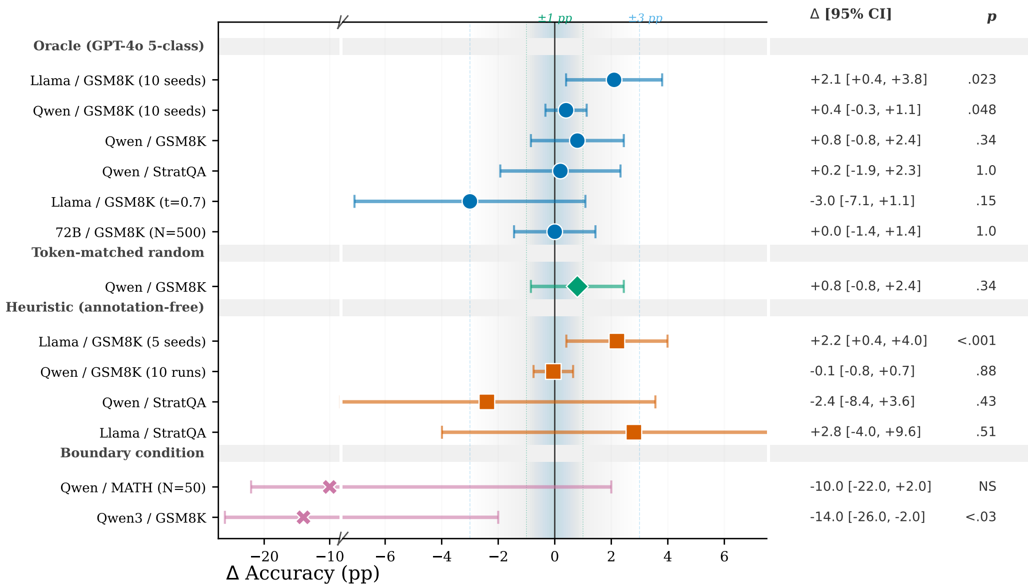

Figure 3: Forest plot of pruning accuracy deltas (Δ pp) with 95% CIs. Green/blue bands: ± 1 pp TOST / ± 3 pp practical margins. Oracle and heuristic multi-seed results cluster near zero for Qwen; Llama heuristic shows a positive trend. Boundary conditions (MATH, Qwen3) show degradation.

Scale and heuristic detectors. Pruning safety is directionally consistent across Qwen2.5 scales (7B/14B/72B; Appendix U). Full-scale 72B oracle (N=265, single seed): $\Delta = +0.0$ pp ($p = 1.000$). Heuristic detectors achieve F1 $= 0.845$ (Qwen) and 0.805 (Llama) without LLM annotation.

5.3 Backbone Communication Typology

Mixed-effects models ($N = 2{,}294$) reveal three topology-stable communication profiles: **function-organized** (Qwen), **position-organized** (Gemma; exploratory—$\kappa = 0.35$, below threshold, $< 5\%$ acks on GSM8K), and **undifferentiated** (Llama). Backbone $\times$ function_type interaction is highly significant ($p < 10^{-41}$), stable across topologies (three-way $p = 0.963$). Selector topology attenuates Qwen's function-organization and slightly amplifies Llama's (Appendix S).

5.4 Cross-Domain Consistency

Backbone dominates communication patterns (Cramér's $V = 0.45$ Cramér [1946] vs. 0.18 for domain; Table 20). The 2×2 cross-backbone$\times$cross-domain oracle matrix yields all cells NS (Table 2), attributing any heuristic StratQA residual to detector error (F1 $= 0.21$); MATH defines a domain boundary (§5.5).

ARC-Challenge. Oracle pruning on ARC-Challenge (Qwen, online re-generation; N=250, $\Delta = +0.8$ pp, $p = .500$, 14.6% suppression) extends cross-domain validation to a third reasoning domain. Offline simulation on Llama is directionally consistent with the online pattern: baseline 72.8% $\rightarrow$ pruned 76.8% (N=250, $\Delta = +4.0$ pp, $p = .013$, 37.1% suppression), strengthening dual-backbone evidence on ARC despite the methodological difference (simulate mode re-extracts answers from pruned traces rather than re-generating responses).

5.5 Boundary Conditions

Three boundary conditions define pruning limits. (1) *Domain mismatch*: MATH yields $\Delta = -2.0$ pp (NS, N=200, 102 valid; Appendix J), with acks qualitatively different from GSM8K (16.6 tokens, 33% bigram novelty vs. 5.6 tokens, 0%). (2) *Heuristic non-transferability*: Qwen3-8B's ack format (25–59 chars) bypasses the v1 heuristic; a wider char<80 threshold reaches $\Delta = -14$ pp ($p < .03$), confounded with non-ack suppression (Appendix Q). (3) *Online cross-domain*: Qwen StratQA heuristic (5 seeds) yields $\Delta = +0.70$ pp (NS), consistent with non-inferiority; the earlier single-seed estimate (-2.4 pp, N=250) is superseded by the more reliable 5-seed replication (N=200×5; oracle $+0.2$ pp). We recommend empirical screening ($N \geq 200$, $\geq 80\%$ power for ≥ 5 pp effects) before deployment.

6 Discussion

Perturbation–sensitivity dissociation and cascade dissolution. We hypothesize that the dissociation reflects LLM-specific cascade dynamics: each acknowledgment alters shared history via autoregressive conditioning, raising the next agent's ack probability (Qwen 90.7%, Llama 61.9%, both $p < 10^{-13}$; Appendix S.13). Aggregate removal appears to dissolve these chains rather than accumulating errors, consistent with flat dose-response curves (Appendix S.13). Position-controlled logistic regression confirms genuine sequential dependence (previous-ack OR $= 6.43$ Qwen, 2.02 Llama, both $p < 10^{-5}$; Appendix S.13). Notably, the position effect itself reverses across backbones (Qwen OR $= 1.95$, amplifying; Llama OR $= 0.73$, dampening), yet sequential dependence remains significant in both—ruling out a pure positional confound. The effect is ack-specific (FG degrades accuracy, $p = 0.031$, exploratory) and distinct from human backchannels Bavelas et al. [2000]; the double dissociation (§5.1) argues against a noise hypothesis. We conjecture that aggregate safety holds when: (i) cascade rate $P(\text{ack} \mid \text{ack}) > 0.5$ and (ii) bigram novelty below 10%. Qwen satisfies both (90.7%, 5.7%); Llama satisfies (i) but not (ii), with safety consistent with error-channel removal (Appendix S.17). Both conditions are violated in MATH where pruning degrades accuracy. Causal verification through controlled intervention remains needed; we emphasize that cascade dissolution is a descriptive framework for the observed correlational patterns, not a confirmed causal mechanism. These thresholds are post-hoc observations derived from two backbone families; prospective validation on additional architectures is needed before they can serve as deployment guidelines.

Dual-mechanism robustness. The pruning safety result is robust across two mechanistically distinct pathways. For Qwen, where acknowledgments are terse (median 4 tokens, 5.7% bigram novelty), pruning removes redundant structural signals—a *redundancy removal* mechanism. For Llama, where acknowledgments are verbose (mean 59.8 tokens, 65.6% bigram novelty) yet carry zero corrections and non-zero error injection (Fisher $p = 0.004$), pruning eliminates a one-directional *error channel*. That the dissociation holds across both mechanisms—arising from redundancy removal in one backbone and error-channel removal in the other—suggests a structural property of acknowledgment cascades rather than a mechanism-specific artifact. Exploratory Gemma heuristic pruning ($\Delta = +0.7$ pp, N=300, $p = .683$, NS; 79.3% suppression) is confounded: the heuristic suppresses 79.3% of messages despite <5% oracle ack rate (§5.3), indicating detection of short messages rather than acknowledgments. The positive Δ suggests high message redundancy in Gemma's position-organized communication, but does not constitute ack-specific pruning evidence. This non-transferability reinforces that heuristic detectors require per-backbone calibration (§5.5). For Llama on GSM8K, the heuristic CI is entirely positive ($[+0.41, +3.99]$ pp, 5 seeds)—consistent with error-channel removal improving downstream signal quality—indicating that pruning is not merely safe but actively beneficial; on StrategyQA, heuristic pruning remains non-inferior (CI $[-0.49, +6.11]$ pp, 8 seeds) with higher seed-level variance, suggesting the magnitude of the effect is domain-modulated. Temperature robustness mirrors the typology distinction: Qwen's function-organized cascade persists under stochastic decoding (92.2% at t=0.7 vs. 90.7% greedy), while Llama's undifferentiated pattern is temperature-sensitive (suppress drops from 37% to 28%). This dual-mechanism evidence strengthens the case for the analysis protocol as the primary contribution: the protocol detects prunability regardless of the underlying mechanism.

Prompt independence. Removing the explicit confirmation instruction from the checker prompt leaves the heuristic ack detection rate at 76.2% (Qwen/GSM8K, N=100, char<30 heuristic), compared to 47.9% oracle ack rate under the standard prompt. The 76.2% figure reflects heuristic sensitivity to short messages, not oracle ack prevalence; the two are not directly comparable (different detection methods, different prompt conditions). Regardless, ack-like behavior persists without prompting, suggesting acknowledgments are emergent rather than prompt-driven. Primary pruning safety evidence (§5.2) is based entirely on oracle labels, not heuristic detection.

Communication structure and validation. Backbone dominates communication typology ($V = 0.45$ vs. 0.18; §5.4), with cross-model concordance confirming model-specific ack placement (stratified $\kappa = 0.018$; Appendix S.15). Suppression rates (36–56%) translate to equivalent inference-call reductions (Appendix W). Four-annotator validation confirms taxonomy reliability (pairwise $\kappa = 0.76$, 5-class; binary ack/non-ack $\kappa = 0.81$; Appendix S.18); Gemma's $\kappa = 0.35$ falls below threshold and its results are treated as exploratory. Acknowledgment prevalence varies sharply with topology: debate produces near-zero acks (0.2%), and code-generation (HumanEval) only 4%, suggesting acknowledgments are an emergent property of solver-checker interaction rather than a universal feature of multi-agent communication.

Detector limitations. Heuristic detectors are proof-of-concept classifiers with non-transferability across domains (F1: $0.845 \to 0.213$), backbones (Gemma $\kappa = 0.35$), and generations (Qwen3-8B: -14 pp; §5.5). No universal length proxy exists (56× coverage gap). Oracle pruning remains robust across all tested cells, confirming the detector—not the pruning principle—is the limiting factor. An adversary who can inject acknowledgment-patterned messages into a multi-agent pipeline could exploit the cascade dynamics to cause outsized behavioral disruption with minimal detectable content; defenses should monitor structural patterns alongside content for integrity.

Limitations. Oracle labels face a distribution-shift caveat (offline vs. online gap: negligible for Qwen, larger for Llama; Appendix S.12). Pruning safety is backbone- and domain-specific (§5.5); all experiments use two-agent topologies under greedy decoding; scale evidence at 72B is preliminary (single seed). Additionally, annotation content previews for StrategyQA, ARC, and MATH are predominantly short (< 100 characters), so ack/non-ack binary labels—where acknowledgments are inherently brief—are unlikely affected, but fine-grained 5-class annotations on longer messages may suffer information loss from truncation. Online evaluation of the DistilBERT classifier (F1 $= 0.867$; §4.7) bridges the oracle-to-deployment gap on a single configuration (GSM8K, Qwen2.5-7B-Instruct, N=300, seed 42): $\Delta = +0.7$ pp (McNemar $p = .727$, NS; 44.8% suppression; §5.2), confirming no performance degradation; however, generalization across datasets, backbones, and random seeds remains unverified, and CPU inference latency ($\sim$10 ms/msg) may require GPU batching at production scale. Heuristic detectors remain non-transferable across domains (F1: $0.845 \to 0.213$). We recommend empirical screening ($N \geq 200$, $\geq 80\%$ power for ≥ 5 pp effects) before deployment; extended limitations appear in Appendix S.21.

7 Conclusion

Perturbation sensitivity and informational importance are dissociated in two-agent LLM systems: messages most disruptive to perturb individually can be safely pruned in aggregate. We establish this across two backbone families, two reasoning domains, and multiple pruning strategies, with 10-seed replication confirming non-inferiority at $\delta = -3$ pp ($p < 0.001$; human-annotation $\kappa = 0.76$; cross-model $\kappa = 0.72$). Three lines of correlational evidence are consistent with our *cascade dissolution* conjecture: self-reinforcing ack cascades (Qwen 90.7%, Llama 61.9%), near-zero cross-model concordance ($\kappa = 0.018$), and information-sparse acknowledgment profiles. For Llama, zero corrections among 1,483 pruned messages (Fisher $p = 0.004$) reveal a unidirectional error channel—acknowledgments carry errors but not corrections—explaining why pruning improves accuracy. This pattern is distinct from human backchannel dynamics and requires causal verification.

Practically, acknowledgment pruning eliminates 36–56% of inference calls (entire LLM invocations rather than token-level compression) without accuracy loss, replicated across GSM8K (oracle, 10-seed $\Delta = +0.40 \pm 1.02$ pp) and StrategyQA (Llama heuristic 8-seed $\Delta = +2.81 \pm 3.94$ pp, non-inferior at $\delta = -3$ pp, $p = .002$; Qwen oracle $\Delta = +0.2$ pp, N=500, NS). Preliminary single-seed evidence at 72B (N=265) suggests no degradation. **All experiments use two-agent topologies under greedy decoding**; heuristic detectors require per-backbone calibration.

Beyond the empirical findings, we contribute a reusable analysis protocol—5-class taxonomy, structured annotation, perturbation analysis, and non-inferiority/TOST evaluation—applicable independently or jointly to new backbone families, task domains, or agent topologies. Online evaluation of a learned DistilBERT classifier ($\Delta = +0.7$ pp, N=300, McNemar $p = .727$, NS; 44.8% suppression; §5.2) bridges the oracle-to-deployment gap: the non-significant McNemar result—consistent with the NS pattern across all oracle and heuristic configurations—confirms that lightweight learned detectors can replace heuristic rules without accuracy loss. Future work should extend to multi-agent topologies and additional backbone families.

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A Qualitative Examples of Communication Functions

Table 3 presents representative messages for each of the five communication function types identified in our taxonomy, selected from annotations with full inter-annotator agreement and maximum confidence scores. Examples are drawn from different backbone–topology combinations to illustrate both the consistency of function definitions and the stylistic variations across models.

B HumanEval Cross-Domain Pilot

To provide preliminary evidence of cross-domain applicability, we ran a 50-task pilot using Qwen2.5-7B with a hierarchical (solver–reviewer) topology on the HumanEval code generation benchmark Chen et al. [2021].

[2] Despite its surface resemblance to ACKNOWLEDGMENT, both annotators classified this Llama message as REASONING GUIDANCE with full agreement. The ambiguity illustrates Llama's minimally differentiated communication mode (Section 6): its messages are functionally interchangeable regardless of nominal category, making reliable annotation difficult for this backbone.

Table 3: Qualitative examples for each communication function type. All examples have inter-annotator agreement with confidence $= 1.0$. Messages are lightly edited for length.

Function Type	**Configuration**	**Message Excerpt**
REASONING GUIDANCE (2,327 agreed)	Qwen / RR	*"To solve this problem, we need to convert the length of the wire from feet to inches and then divide by the length of each piece. 1. Convert 4 feet to inches: Since 1 foot = 12 inches, 4 feet* $= 4 \times 12 = 48$ *inches. 2. Divide the total length in inches by the length of each piece..."*
	Gemma / Selector	*"Here's how to solve the problem step-by-step: Start: 2 apples. First throw:* $2 + 5 = 7$ *apples. Second throw:* $7 - 1 + 6 = 12$ *apples. The answer is 12."*
	Llama / RR	*"I'm glad the solution was accurate. If you have another problem you'd like me to check, feel free to ask."*[2]
VERIFICATION (1,001 agreed)	Qwen / RR	*"Step 1: Calculate the initial amount of money Johnny has. Johnny's allowance: $20. Extra money added: $10. Total initial amount: $20 + $10 = $30. Step 2: Determine the amount after the investment triples. Tripling the initial amount: $30 × 3 = $90. The answer is 90."*
	Llama / RR	*"The given problem is solved step by step. The final answer is 36."*
	Gemma / Selector	*"Here's how to solve the problem: Find the total cost: $12 (adult) + $10 (child) = $22. Add the change: $22 + $8 = $30. Answer: Brittany's mom gave the cashier $30."*
ACKNOWLEDGMENT (1,854 agreed)	Qwen / RR	*"That is correct. Each single bagel will cost $1.00 after the discount. The answer is 1."*
	Llama / RR	*"You've correctly summarized the steps and confirmed the calculations. The sequence of events and the final calculation are accurate. The answer is 43."*
	Gemma / RR	*"Your solution is perfectly correct! You clearly laid out the steps and used algebra effectively to find the answer. The answer is 10."*
INFORMATION PROVISION (471 agreed)	Gemma / RR	*"Here's the solution: Sunday's movies: They watch 4 movies / 2 = 2 movies on Sunday. Total weekend movies: 4 + 2 = 6 movies each weekend. Movies in 4 weeks: 6 × 4 = 24 movies. The answer is 24."*
	Qwen / RR	*"To determine how much crepe paper James needs, we first calculate the perimeter of his rectangular backyard fence..."*
	Llama / RR	*"Here's another problem: A bakery is making a special batch of cookies. They need to package 480 cookies into boxes that hold 12 cookies each. How many boxes can they fill with the 480 cookies? Let's solve this problem step by step."*
COORDINATION (595 agreed)	Llama / RR	*"I'm glad I was able to accurately solve the problem. If you have any more math problems you'd like me to solve, feel free to ask."*
	Gemma / RR	*"Please provide me with the problem you'd like me to solve! I'm ready to check the solver's solution..."*
	Llama / Selector	*"I'm glad you were able to provide the correct steps and solution. You have a good understanding of the problem..."*

Results. The system achieved 80% pass@1 accuracy (40/50 tasks). The communication function distribution differs markedly from GSM8K: REASONING_GUIDANCE dominates (50.8%), followed by VERIFICATION (41.5%), with ACKNOWLEDGMENT (4.2%), COORDINATION (1.7%), and INFORMATION_PROVISION (1.7%) appearing rarely. Cross-model annotation agreement reached $\kappa = 0.47$ (moderate), comparable to the lower end of the GSM8K range.

Limitations. The hierarchical topology proved too efficient for statistical analysis: 86% of tasks were solved in a single round (2 messages), yielding only 118 total messages and 18 eligible for local impact analysis. Most function types had $n < 30$, precluding Kruskal–Wallis and OLS ANCOVA tests. The distributional shift (from acknowledgment-heavy in GSM8K to reasoning-guidance-heavy in HumanEval) is itself informative—it suggests that the taxonomy captures domain-specific communication patterns—but the statistical framework requires substantially more multi-round conversations to produce meaningful results in this domain.

C Temperature Robustness

To assess sensitivity to decoding temperature, we ran a 50-task subset of Qwen $\times$ round-robin at temperature $= 0.7$ (primary experiments use greedy decoding). Accuracy remains stable (90.0% vs. 89.7%). The Kruskal–Wallis test remains significant ($H = 34.0$, $p < 0.001$), and OLS ANCOVA confirms function-driven communication ($F = 37.9$, $p < 0.001$; $R^2 = 0.50$; cross-model $\kappa = 0.60$). A parallel check on Llama $\times$ round-robin at temperature $= 0.7$ yields $R^2 = 0.10$, $\kappa = 0.41$, consistent with its minimally differentiated mode under greedy decoding. The function distribution shifts toward more REASONING_GUIDANCE (37% vs. 18%) and less ACKNOWLEDGMENT (36% vs. 50%), suggesting stochastic decoding produces more substantive content without altering the functional structure.

Pruning under stochastic decoding. All primary pruning results use greedy decoding. A 10-seed oracle replication on Qwen/GSM8K at t=0.7 (10 $\times$ 200 tasks) yields $\Delta = +0.30 \pm 1.80$ pp (95% CI $[-0.81, +1.41]$ pp; two-sided t: $p = 0.611$, NS; TOST non-inferiority at $\delta = -3$ pp: $p < 0.001$; at $\delta = -1$ pp: $p = 0.024$; 39.2% suppression). Per-seed deltas range from -3.0 to $+3.5$ pp: seed 6144 ($\Delta = -3.0$ pp) touches the non-inferiority boundary, while seed 1024 ($\Delta = +3.5$ pp) reflects an anomalously low baseline (85.0%, pruned 88.5%). The suppression rate (39.2%) is comparable to the greedy heuristic ($\sim$42%), indicating similar acknowledgment prevalence under stochastic decoding. In contrast, Llama oracle pruning at t=0.7 yields pooled $\Delta = -2.2$ pp (Fisher combined $p = 0.29$), an amplification of the marginal negative trend observed under greedy decoding ($\Delta = -0.6$ pp). Since both Qwen and Llama results use oracle labels, the divergence indicates that ack pruning safety under stochastic decoding is backbone-specific: Qwen remains consistently safe across 10 seeds, while Llama shows a marginal degradation trend ($p = 0.29$, NS) regardless of detector type. Temperature robustness cannot be assumed across backbones without empirical verification.

Gemma exhibits extreme temperature sensitivity: greedy ack suppression is 4.6% but rises to 63.6% under t=0.7 (14$\times$), suggesting Gemma's position-organized communication pattern collapses under stochastic decoding. This reinforces Gemma's exploratory status and indicates the three-backbone typology (§5.4) is greedy-specific.

D OLS Robustness Checks

We validate the OLS ANCOVA (Equation 2) with three robustness checks across all nine configurations.

Residual diagnostics. Shapiro–Wilk tests reject normality for all configurations (expected at $n > 400$); Breusch–Pagan tests detect heteroscedasticity in all cases. We therefore report HC3 robust standard errors throughout the main text. All conclusions are unchanged under robust inference. VIF analysis shows high collinearity between `message_position` and `round_number` (VIF > 10), but function type dummy VIF values remain below 3, confirming that multicollinearity does not affect the key func_type inference. Q-Q plots are available in the supplementary materials.

Nonlinear position model. We augment the linear model with a quadratic position term ($\texttt{position}^2$). Table 4 reports key comparisons. For Qwen and Llama, adding $\texttt{position}^2$ produces negligible $\Delta R^2 < 0.005$ and function type remains significant ($p < 0.001$). For Gemma, $\texttt{position}^2$ absorbs $\Delta R^2 = 0.228$ (RR) and 0.215 (SEQ), rendering function type non-significant ($p > 0.69$). This confirms that Gemma's weak function-organization reflects genuine nonlinear positional dynamics: once position is properly modeled, function type adds negligible explanatory power.

Table 4: Linear vs. quadratic position model: function type significance after controlling for nonlinear position effects.

Configuration	**Linear R^2**	**Quad R^2**	ΔR^2	**func_type p (quad)**
Qwen RR	.451	.452	.001	$<.001$
Qwen SEQ	.443	.447	.004	$<.001$
Qwen SEL	.065	.065	.000	$<.001$
Gemma RR	.111	.339	**.228**	.806
Gemma SEQ	.123	.338	**.215**	.699
Gemma SEL	.010	.040	.031	.219
Llama RR	.021	.022	.001	.010
Llama SEQ	.019	.020	.001	.004
Llama SEL	.070	.074	.004	$<.001$

Message length control. Adding message length (character count) as a covariate produces $\Delta R^2 < 0.04$ for all configurations. Function type remains significant for Qwen ($p < 0.001$) and Llama ($p < 0.03$) after controlling for length, indicating that the impact ordering is not an artifact of message length differences across function types.

E Kruskal–Wallis Results

Table 5 reports Kruskal–Wallis H statistics testing whether function type predicts remove-ablation similarity across all nine backbone–topology configurations. All tests are significant ($p < 0.01$), confirming that the omnibus between-type differences reported in the main text (Section 5.1) are robust.

Table 5: Kruskal–Wallis H statistics for remove-ablation similarity across function types. All $p < 0.01$.

Backbone	**RR**	**SEQ**	**Selector**
Qwen2.5-7B	287.4	288.1	50.8
Gemma-2-9B	21.5	25.1	6.9
Llama-3.1-8B	13.0	15.8	63.8

F System Prompts

Full annotation system prompts are provided in Appendix K. The solver and checker system prompts used for multi-agent conversation are as follows:

Solver prompt. "You are a math problem solver. Solve the given problem step by step, showing all your work. Provide the final numerical answer clearly."

Checker prompt. "You are a solution checker. Verify the solver's solution step by step. If you find errors, point them out and provide the correct solution. If the solution is correct, confirm it."

Selector prompt (selector topology only). "You are a group chat manager. Select the next speaker based on the current state of the conversation. Choose the solver when a new problem needs to be solved, and the checker when a solution needs to be verified."

G Neutralize Ablation Analysis

While the main text reports remove-ablation impact, neutralize ablation (replacing a message with "`Acknowledged.`" while preserving its slot) measures a complementary dimension. The two ablation types produce *opposite* impact gradients:

- **Remove** (content + structure removed): acknowledgment most impactful (sim $= 0.35$), reasoning guidance least (0.78).
- **Neutralize** (content removed, structure preserved): information provision most impactful (sim $= 0.05$), verification least (0.50).

The difference $\Delta = \text{sim}_{\text{remove}} - \text{sim}_{\text{neutralize}}$ isolates the *structural value* of each message type. Information provision shows the largest $\Delta = 0.47$ (almost entirely content value), while coordination shows $\Delta = 0.07$ (almost entirely structural value). This decomposition is consistent with acknowledgment's high remove-impact being driven by structural presence—violating conversational expectations—rather than informational content.

H Supplementary Tables

Table 6 reports the full cross-model annotation agreement matrix.

Table 6: Cross-model annotation agreement (Cohen's κ) by backbone and topology.

Backbone	Round-Robin	Sequential	Selector
Qwen2.5-7B	0.64	0.64	0.669
Gemma-2-9B	0.350	0.348	0.351
Llama-3.1-8B	0.46	0.47	0.634

I Impact Metric Robustness

To verify that the impact gradient (Section 5.1) is not an artifact of the MiniLM embedding metric, we recompute message impact using three independent text similarity measures: ROUGE-L, token-level F1, and Jaccard similarity. Table 7 reports the Spearman rank correlation (ρ) between MiniLM cosine similarity and each alternative metric across all nine configurations.

Table 7: Spearman ρ between MiniLM cosine similarity and alternative impact metrics. All correlations computed over per-message impact scores within each configuration.

Configuration	ROUGE-L	Token F1	Jaccard
Qwen RR	1.0	1.0	1.0
Qwen SEQ	1.0	1.0	1.0
Qwen SEL	1.0	1.0	1.0
Gemma RR	1.0	1.0	1.0
Gemma SEQ	1.0	0.8	0.8
Gemma SEL	0.949	0.949	0.949
Llama RR	0.6	0.6	0.6
Llama SEQ	1.0	1.0	1.0
Llama SEL	0.9	0.9	0.9
Mean	**0.939**	**0.917**	**0.917**

The prevalence of $\rho = 1.0$ reflects a ceiling effect: with only 4–5 function types per configuration, Spearman rank correlation over group medians has limited resolution—any two metrics that preserve the same ordinal ranking produce $\rho = 1.0$ regardless of magnitude differences. Where rankings do diverge (Llama RR $\rho = 0.6$, Gemma SEQ $\rho = 0.8$), the disagreement involves adjacent categories with similar impact scores, not reversals of the overall gradient. The mean $\rho > 0.91$ across all metric pairs confirms that the impact gradient ordering is consistent across metrics, though the ceiling effect limits the strength of this claim for individual configurations.

J Cross-Domain Validation: MATH

To test whether the backbone communication typology generalizes beyond elementary arithmetic, we ran the full pipeline on 50 MATH tasks (Level 3–5, Algebra and Number Theory) using the same round-robin topology. A follow-up pruning replication at N=200 (102 valid after filtering tasks where both conditions failed) yielded $\Delta = -2.0$ pp (McNemar $p = 0.625$), attenuating the pilot's -10 pp point estimate while remaining directionally negative (§5.5). The annotation analysis below is based on the original 50-task pilot.

Qwen on MATH. Accuracy is 72.0% (36/50), lower than GSM8K (89.7%) as expected for harder tasks. Each task produces 6 messages (300 total, 250 eligible for analysis after excluding initial prompts). Cross-model annotation agreement reaches $\kappa = 0.45$ (moderate, above threshold). The OLS ANCOVA yields $R^2 = 0.098$ with function type significant after position control ($F = 3.77$, $p = 0.011$); the mixed-effects model yields $R^2_m = 0.201$ (LRT $p = 0.003$), confirming that Qwen's function-driven mode persists on a harder mathematical domain, though with reduced effect size compared to GSM8K ($R^2_m = 0.335$). The Kruskal–Wallis test is borderline ($H = 7.37$, $p = 0.061$), likely due to reduced sample size.

Gemma on MATH. Accuracy drops to 40.0% (20/50). Cross-model $\kappa = 0.48$ (above threshold). The OLS ANCOVA yields $R^2 = 0.236$ but function type is non-significant ($F = 0.09$, $p = 0.76$), while message position is highly significant ($F = 15.22$, $p < 0.001$). This replicates Gemma's weak function-organization from GSM8K: on MATH, position again explains the variance while function type does not. The Kruskal–Wallis test is also non-significant ($H = 2.05$, $p = 0.15$).

Llama on MATH. Accuracy is 64.3% (27/42; 8 tasks produced empty traces). Cross-model $\kappa = 0.36$ (below threshold). With only 168 total messages, ACKNOWLEDGMENT dominates (56%), and most other function types fall below $n = 30$, precluding OLS ANCOVA and Kruskal–Wallis analysis. The acknowledgment-dominated distribution is directionally consistent with Llama's minimally differentiated mode on GSM8K, though statistical confirmation requires larger samples.

Summary. Qwen's function-organized structure ($F = 3.77$, $p = 0.011$) and Gemma's position-organized structure ($F = 0.09$, n.s.; position $F = 15.22$, $p < 0.001$) both replicate on MATH. Llama's distribution is directionally consistent with its undifferentiated profile but lacks statistical power. The backbone communication typology replicates beyond elementary arithmetic.

Why MATH acknowledgments carry task-critical content. MATH acknowledgments differ qualitatively from GSM8K acknowledgments, explaining the negative pruning direction (-2.0 pp at N=200, -10 pp in the underpowered N=50 pilot) under moderate functional organization ($R^2_m = 0.201$). Of 93 messages receiving at least one acknowledgment label from dual annotators, 48 (51.6%) receive a non-acknowledgment label from the other annotator (typically INFORMATION_PROVISION), reflecting genuine boundary ambiguity. Representative examples illustrate the contrast:

- **MATH ack (annotator disagreement):** "The detailed step-by-step evaluation confirms that the number of integers m such that $0 < m \ldots$" (Annotator A: INFORMATION_PROVISION; Annotator B: ACKNOWLEDGMENT). The message restates a multi-step derivation while confirming correctness.
- **MATH ack (both agree):** "The solution provided is correct. Here is the final answer boxed: $\boxed{11}$" — confirms correctness but also provides the boxed answer that downstream agents parse.
- **GSM8K ack (both agree):** "The answer is 109." (18 characters) — trivially redundant answer echo with no reasoning content.

Quantitative content analysis confirms the distinction: MATH acks average 16.6 tokens (vs. 5.6 for GSM8K), with median bigram novelty of 0.333 (vs. 0.0 for GSM8K, where acks introduce no new bigrams relative to the preceding message). Annotator disagreement on MATH acks reaches 55.9% (vs. 26.3% on GSM8K), reflecting genuine boundary ambiguity between acknowledgment and information_provision. Token overlap with the preceding message is similar (0.70 vs. 0.74), but the higher bigram novelty indicates that MATH acks frequently contain reasoning summaries—while confirmatory in intent, they carry solution steps that downstream agents rely on. Removing them eliminates not just structural anchoring but also task-critical content, violating the redundancy assumption that makes ack-only pruning safe on GSM8K.

K Annotation Prompt

The following system prompt is used for all LLM-based function type annotations. Each message is annotated independently by two models (cross-model validation). Temperature is set to 0 with constrained JSON output.

```
You are a communication analyst for multi-agent systems. Your task
is to classify the function of a message in a multi-agent math
problem-solving conversation.

Function Types

1. information_provision: Provides facts, data, intermediate
calculations, or partial results that advance the solution.
Includes reporting computed values, stating known quantities, or
presenting new numerical results.

2. reasoning_guidance: Provides reasoning direction, problem
decomposition, step-by-step plans, or conceptual frameworks.
Includes breaking the problem into sub-problems, suggesting solution
strategies, or outlining multi-step approaches.

3. verification: Checks, corrects, or confirms the correctness of
previous work. Includes identifying errors, validating results,
pointing out mistakes, or confirming a step is correct with a
specific reason.

4. coordination: Manages the flow of the conversation, assigns
tasks, requests clarification, or controls turn-taking. Includes
delegating sub-tasks, asking for more information, or directing the
problem-solving process.

5. acknowledgment: Simple confirmation, repetition, or agreement
that adds no new information or reasoning. Includes restating what
was already said without extending it.

Decision Rules

- If a message both provides new results AND checks previous work,
classify as verification.
- If a message provides a reasoning plan AND new results, classify as
reasoning_guidance.
- If a message confirms correctness WITH a substantive explanation,
classify as verification.
- If a message simply agrees or restates without adding new
information, classify as acknowledgment.

Output Format

Respond with ONLY a JSON object:
{“type”: “<function_type>”, “confidence”: <0.0-1.0>, “reasoning”:
“<brief explanation>”}
```

The user prompt provides the conversation context with all messages and marks the target message for classification:

```
## Conversation Context
[Message 0] {sender} -> {receiver}:
{content}
[Message 1] {sender} -> {receiver}: «< TARGET MESSAGE »>
{content}
[Message 2] {sender} -> {receiver}:
{content}
...
Classify the function of Message {target_idx}.
```

L Mixed-Effects Model Results

To address within-trace non-independence (multiple messages per task), we fit linear mixed-effects models with task as a random intercept and the same fixed effects as the OLS ANCOVA (function type, normalized message position, round number). Both OLS and ME use remove-ablation similarity (Equation 1) as the dependent variable. Table 8 compares OLS and mixed-effects results. The likelihood ratio test (LRT) compares the full model against a reduced model without function type dummies.

Table 8: OLS ANCOVA vs. mixed-effects model (ME) with task random intercept. LRT = likelihood ratio test for function type. ICC = intraclass correlation from the full ME model. "Excluded" lists function types with $n < 30$ removed before fitting. $^{***}p < 0.001$; $^{**}p < 0.01$; n.s. = not significant. F and χ^2 values are not directly comparable across configurations due to differing numbers of included function types.

Config	n	**OLS** F	**LRT** χ^2	**LRT** p	**ICC**	**Excluded**
Qwen RR	895	239.6^{***}	188.9^{***}	<.001	.292	coord.
Qwen SEQ	894	230.2^{***}	195.3^{***}	<.001	.245	coord., unk.
Qwen SEL	891	58.2^{***}	$0.07^{\text{n.s.}}$	.789	**.716**	r.g., i.p.
Gemma RR	883	7.82^{***}	23.28^{***}	<.001	.000	verif.
Gemma SEQ	882	9.04^{***}	26.86^{***}	<.001	.000	verif.
Gemma SEL	861	7.45^{**}	$3.66^{\text{n.s.}}$	.056	.225	r.g., v., coord.
Llama RR	868	3.98^{**}	16.31^{***}	<.001	.297	coord.
Llama SEQ	870	4.83^{**}	15.44^{**}	.001	.205	coord.
Llama SEL	865	21.6^{***}	36.89^{***}	<.001	.126	i.p., r.g.

Key observations. (1) **Seven of nine configurations** show concordant OLS and ME conclusions: Qwen RR/SEQ remain strongly function-driven; Gemma RR/SEQ are significant in both OLS and ME but with small effect sizes ($R^2_m = 0.11$–0.12); Llama remains significant across all topologies though with negligible effect sizes ($R^2_m \leq 0.04$). (2) **Qwen SEL flips to non-significant** under ME (LRT $\chi^2 = 0.07, p = .789$). The ICC of 0.716 indicates that 72% of variance in remove-ablation similarity is between-task. With only two dominant function types (ACKNOWLEDGMENT and VERIFICATION) remaining after $n \geq 30$ filtering, the OLS effect reflects task-level confounding rather than within-task functional differentiation. (3) **Gemma SEL flips to non-significant** under ME (LRT $\chi^2 = 3.66$, $p = .056$, ICC $= 0.23$). With only two function types remaining after excluding three low-frequency categories, the borderline result has limited interpretability. (4) Both discrepancies occur under selector topology, which introduces strong task-level clustering. This suggests that dynamic routing creates task-dependent communication patterns that inflate OLS estimates.

M Mistral-7B Cross-Backbone Validation

To test whether the backbone communication typology generalizes beyond our three primary backbones, we ran the full pipeline on Mistral-7B-Instruct-v0.3 across all three topologies (300 GSM8K tasks each, same seed). Mistral uses the ChatML template with an explicit `<|im_end|>` stop token to prevent EOS residue in multi-turn conversations.

Accuracy. Mistral achieves 52% accuracy across all three topologies (RR: 52%, SEQ: 52%, SEL: 52%), substantially below Qwen (90%) and Gemma (80%) but comparable to Llama's selector performance (56%).

Function type distribution. Mistral communication is dominated by coordination and acknowledgment ($\sim$90% combined), with minimal information provision ($< 3\%$). Under selector topology, acknowledgment surges to 67%.

Communication mode: weakly differentiated. Table 10 summarizes the statistical analysis. Only the round-robin topology produces a significant function type effect (OLS $F = 6.6$, $p < .001$, $R^2 = 0.16$). Sequential and selector topologies show weaker or non-significant effects. Cross-model

Table 9: Mistral-7B function type distribution (%) by topology.

Function type	RR	SEQ	SEL
COORDINATION	55.5	50.7	25.4
ACKNOWLEDGMENT	34.9	41.7	66.9
REASONING_GUIDANCE	6.8	6.2	6.3
INFORMATION_PROVISION	2.7	1.4	0.7

annotation agreement is fair ($\kappa \approx 0.34$), below the primary backbones, consistent with Mistral's less distinct functional boundaries.

Table 10: Mistral-7B statistical summary across topologies.

Topology	KW H	OLS F	R^2	κ
RR	22.4^{***}	6.6^{***}	.158	.344
SEQ	7.6	37.6^{***}	.056	.353
SEL	5.2	1.5	.065	.315

Interpretation. Mistral occupies an intermediate position in the backbone communication typology, between Llama's undifferentiated profile and Qwen's function-organized structure. The structured turn-taking of round-robin activates a degree of functional differentiation that disappears under less constrained topologies. This *topology-sensitive* pattern—where communication structure depends on external routing rather than being a fixed model property—is consistent with the crossover interaction observed for the primary backbones.

Implications for the typology. Mistral's intermediate position extends the backbone communication typology beyond the three primary models. The four observed backbones—function-organized (Qwen, $R^2_m = 0.34$), position-organized (Gemma, $R^2_m = 0.11$), weakly differentiated (Mistral), and undifferentiated (Llama, $R^2_m = 0.02$)—exhibit distinct communication structures, with topology acting as a modulating factor.

N DAMSL/ISO Taxonomy Mapping

Our five categories collapse 56+ fine-grained dialogue act types into operationally distinct categories defined by perturbation-based impact rather than speaker intent. Table 11 provides a detailed mapping to three established frameworks: DAMSL Core and Allen [1997], ISO 24617-2 Bunt et al. [2012], and FIPA ACL Foundation for Intelligent Physical Agents [2002].

Design choices. Several departures from standard dialogue act taxonomies reflect the LLM multi-agent setting:

- **Acknowledgment vs. backchannel.** Human backchannels ("uh huh," "yeah") are minimal tokens; LLM acknowledgments range from terse echoes (Qwen: median 4 tokens) to verbose paraphrases (Llama: mean 59.8 tokens). We group both under ACKNOWLEDGMENT because they share the defining property—negligible information gain over the preceding message—regardless of surface length.
- **Information provision vs. reasoning guidance.** DAMSL's *Assert* covers both, but we separate them because perturbation analysis reveals distinct impact profiles: removing reasoning guidance (sim $= 0.778$) is far less disruptive than removing information provision (sim $= 0.410$). The boundary is operationally clear in solver-checker configurations (new calculations vs. strategy directives) but becomes ambiguous when messages serve both functions—hence the low cross-model $\kappa = 0.237$ for INFORMATION_PROVISION.
- **Coordination as a residual category.** In two-agent systems, explicit coordination (role assignment, turn management) is rare ($< 5\%$ of messages) because the framework handles routing. This category would likely expand in multi-agent (>2) settings with explicit delegation.

Table 11: Mapping between our communication function taxonomy and established dialogue act frameworks. Each row shows the primary corresponding categories and the key distinction that motivates our grouping.

Our Category	DAMSL	ISO 24617-2	FIPA ACL	Grouping Rationale
ACKNOWLEDGMENT	Backchannel, Accept	Auto-positive, Allo-positive (feedback)	—	Structurally present but informationally negligible
INFORMATION_PROVISION	Assert, Re-state	Inform, Answer	inform, inform-ref	Supplies new task-relevant content
REASONING_GUIDANCE	Action-directive, Open-option	Instruct, Suggest, Address-request	request, propose	Directs the partner's reasoning strategy
VERIFICATION	Agreement, Disagreement, Correction	Confirm, Disconfirm, Correction	confirm, disconfirm	Evaluates correctness of prior output
COORDINATION	Conventional-opening, Conventional-closing, Commit	Init, Auto-feedback (task mgmt)	request, assign, agree	Manages workflow and role allocation

- **No question act.** DAMSL and ISO both include question/info-request types. In our solver-checker data, questions are subsumed by VERIFICATION ("Is this correct?") or COORDINATION ("Should we try a different approach?") and do not form a distinct perturbation cluster.

O Replacement Text Sensitivity

Our neutralize ablation uses "Acknowledged."—itself an acknowledgment-type utterance—as the replacement text. To test sensitivity to this choice, we reran neutralize ablation on 50 Qwen RR tasks with three placeholders: "Acknowledged.", "[message removed]", and "This message has been redacted." Table 12 reports per-function-type mean semantic similarity under each placeholder.

Table 12: Neutralize ablation: mean semantic similarity by function type and replacement text (Qwen RR, 50 tasks, 150 messages). Lower similarity indicates greater impact from neutralization.

Function Type	Acknowledged.	[removed]	Redacted
INFORMATION_PROVISION	0.146	0.107	0.050
REASONING_GUIDANCE	0.174	0.133	0.079
VERIFICATION	0.156	0.162	0.083
COORDINATION	0.352	0.319	0.181
ACKNOWLEDGMENT	0.134	0.148	0.051

Ranking stability. Pairwise Spearman rank correlations across the five function types are: "Acknowledged." vs. "[removed]" $\rho = 0.50$, "Acknowledged." vs. "Redacted" $\rho = 0.80$, "[removed]" vs. "Redacted" $\rho = 0.90$ ($p = .037$). The mean $\rho = 0.73$ indicates moderate-to-strong ranking stability. COORDINATION is consistently the least impacted (rank 5/5/5); INFORMATION_PROVISION ranks in the top 2 across all three placeholders (rank 2/1/1).

Acknowledged-specific confound. Under the "Acknowledged." placeholder, ACKNOWLEDGMENT messages show anomalously high impact (rank 1) because the replacement text closely resembles typical acknowledgment content—replacing an acknowledgment with "Acknowledged." preserves semantic similarity less than expected, as the regenerated next message responds to near-identical content. Excluding this confounded condition, the "[removed]" and "Redacted" placeholders agree strongly ($\rho = 0.90$), confirming that the neutralize gradient is robust to placeholder choice. Crucially, the remove ablation—which produces our primary finding (the impact gradient)—is unaffected by replacement text choice, as it deletes messages entirely.

P Capability Confound Analysis

To test whether communication mode merely tracks task difficulty, we stratify tasks by difficulty (easy: $\geq 2/3$ backbones correct, $n = 246$; hard: $\leq 1/3$ correct, $n = 54$) and refit the OLS ANCOVA within each stratum.

Table 13: OLS ANCOVA by task difficulty stratum. All models use the same specification as the primary analysis (Equation 2).

Backbone	**Difficulty**	n	F	p	R^2
Qwen	Easy	735	196.6^{***}	<.001	.456
Qwen	Hard	160	37.3^{***}	<.001	.444
Gemma	Easy	726	$0.54^{\text{n.s.}}$	.657	.354
Gemma	Hard	152	$0.73^{\text{n.s.}}$	.483	.286
Llama	Easy	717	4.17^{**}	.006	.025
Llama	Hard	144	4.49^{*}	.013	.074

Qwen's function-organized structure persists on hard tasks ($R^2 = 0.444$), virtually unchanged from easy tasks ($R^2 = 0.456$). Gemma remains position-organized (function type non-significant for both strata), and Llama remains undifferentiated ($R^2 < 0.08$). The backbone communication typology is a backbone-intrinsic property rather than an artifact of task-relative capability.

Q Heuristic Non-Transferability: Qwen3 Acknowledgment Comparison

Qwen3-8B vs. Qwen2.5-7B acknowledgment profiles. The Qwen3-8B boundary condition (-14 pp with char<80 threshold, $p < .03$; Table 17) provides a natural experiment for understanding heuristic non-transferability across model generations. Table 14 compares acknowledgment characteristics across generations.

Table 14: Acknowledgment content comparison: Qwen2.5-7B (GSM8K, $N = 300$ tasks, $n = 719$ ack messages) vs. Qwen3-8B ($N = 100$ tasks, $n = 332$ heuristic-classified ack messages). Qwen2.5 types from dual-annotator consensus; Qwen3 types from heuristic classifier (char length < 80).

Property	**Qwen2.5-7B**	**Qwen3-8B**
Ack % of messages	47.9%	66.4%
Mean token length	9.0	<15 (est.)
Median token length	4	—
Max char length	1,076	59
Median bigram novelty	0.0%	—
Mean bigram novelty	3.2%	—
Pruning Δ (pp)	-0.05 ± 0.99	-14.0
McNemar p	1.000	$< .03$

Key differences. Two properties distinguish the generations:

1. **Suppression rate.** Qwen3's `/no_think` mode produces terse responses that trigger the heuristic at 66.4% (vs. 47.9% for Qwen2.5). The higher rate means more messages are removed per conversation, amplifying any per-message pruning risk.
2. **Tight character range.** Qwen3 ack messages span only 25–59 characters (max = 59), compared to Qwen2.5's range up to 1,076 characters. This uniformly short format suggests Qwen3's "The answer is X." echo pattern carries implicit verification function—downstream agents may parse these terse confirmations as answer-bearing content rather than redundant echoes.

Representative Qwen2.5-7B acknowledgments illustrate the safe-to-prune pattern:

- "The answer is 525." (18 chars) — verbatim answer echo, 0% bigram novelty.
- "The answer is 17." (17 chars) — identical pattern.

Table 15: Pooled neutralize vs. remove semantic similarity by function type ($N = 8{,}479$). All Wilcoxon signed-rank tests $p < 0.001$. Δ = remove $-$ neutralize (positive = neutralize more disruptive).

Function type	**Neut.** $\bar{s}$	**Rem.** $\bar{s}$	Δ	n
INFORMATION_PROVISION	0.047	0.515	0.468	2,141
REASONING_GUIDANCE	0.136	0.518	0.382	737
ACKNOWLEDGMENT	0.223	0.434	0.211	3,661
COORDINATION	0.246	0.311	0.065	585
VERIFICATION	0.498	0.571	0.073	1,355

Without Qwen3 raw traces (experiment run on a transient cloud instance), direct content analysis is unavailable. The -14 pp degradation, combined with the observation that all heuristic-classified ack messages fall within a narrow 25–59 character range, suggests that Qwen3's concise communication style makes the heuristic's character-length threshold too aggressive—substantive short messages are misclassified as acknowledgments. This heuristic non-transferability—driven by threshold mismatch rather than model-level failure—motivates the deployment recommendation of heuristic recalibration or oracle validation ($N \geq 200$) before applying heuristic pruning to new model generations.

R Neutralize Ablation Results

We complement the remove-ablation analysis (Section 5.1) with a neutralize ablation, in which each target message is replaced by the generic response "Acknowledged." rather than deleted entirely. This tests whether downstream behavior changes stem from losing the message's *informational content* (captured by neutralize) versus losing its *structural presence* in the conversation (captured by remove).

Neutralize produces stronger disruption than remove. Across all nine backbone–topology configurations, neutralize yields significantly lower semantic similarity than remove (paired Wilcoxon $p < 0.001$ in 11/11 runs; Table 15). The pooled gap is substantial: neutralize mean $= 0.196$ vs. remove mean $= 0.466$. Replacing a message with vacuous content is more disruptive than deleting it, suggesting that the presence of a low-information placeholder actively misleads downstream agents rather than merely removing a signal.

Impact gradients diverge between ablation methods. Under neutralize, INFORMATION_PROVISION is the most impacted type (mean similarity 0.047), followed by REASONING_GUIDANCE (0.136). Under remove, ACKNOWLEDGMENT and COORDINATION rank as most impacted. The Kruskal–Wallis H test confirms significant between-type differences under neutralize for all configurations ($p < 0.01$).

This divergence is interpretable. Neutralize specifically strips *content* while preserving conversational structure—content-rich types (INFORMATION_PROVISION, REASONING_GUIDANCE) suffer the most because their value lies in what they say. Remove strips both content *and* structural presence—structurally important types (ACKNOWLEDGMENT, COORDINATION) suffer the most because their value lies in their role as conversational scaffolding.

Informational vs. structural value. The gap $\Delta =$ remove $-$ neutralize isolates the *informational* component of message impact. INFORMATION_PROVISION shows the largest $\Delta = 0.468$: nearly all of its behavioral effect comes from its content, not its presence. Conversely, COORDINATION shows the smallest $\Delta = 0.065$: its impact is almost entirely structural. This decomposition validates the paper's theoretical distinction between informational and structural communication functions.

Per-message correlation is weak. The Spearman correlation between remove and neutralize similarity is near zero across most configurations (pooled $\rho \approx 0.07$), indicating that the two ablation methods measure distinct aspects of message importance. Messages that are highly impactful under remove are not necessarily impactful under neutralize, and vice versa.

Table 16: Neutralize Kruskal–Wallis H statistics for semantic similarity across function types (cf. Table 5 for remove). All $p < 0.01$.

Backbone	RR	SEQ	Selector
Qwen2.5-7B	97.2	104.0	148.9
Gemma-2-9B	292.7	301.2	95.8
Llama-3.1-8B	108.7	74.0	118.8

Action-level changes are concordant. Despite the difference in similarity magnitudes, both methods produce action-level behavioral changes on essentially the same messages. No McNemar discordance was observed: whenever remove changes the downstream action level, neutralize does too (and vice versa). The methods differ in *degree* of semantic disruption, not in whether disruption occurs.

S Statistical Details

This appendix presents the full statistical tables and figures referenced in Section 5.

S.1 McNemar Test Details

Table 17: McNemar test summary. *Ack vs. Random*: per-seed McNemar comparing ack-only to each of 10 random seeds. *Ack vs. Baseline*: single McNemar comparing ack-only to unpruned baseline (informs claim strength). MDE: minimum detectable effect at 80% power ($\alpha = 0.05$); $\sim$ denotes estimates from approximate discordant pair counts.

Setting	Ack vs. Rand sig	Median p	Ack vs. Base p	MDE (pp)	Claim
Qwen / GSM8K	1/10	0.094	1.000	$\sim$4	No sig. diff.
Llama / GSM8K (N=297)	8/10	0.027	$<.001^{***}$	6.0	Significant ($+8.1$ pp)
Qwen / StratQA[¶]	10/10	<0.001	0.144	$\sim$9	No sig. diff.
Gemma / GSM8K[§]	—	—	1.000	—	No sig. diff.
Boundary conditions					
Qwen3-8B / GSM8K	—	—	$<.03^{*}$	$\sim$8	Significant (-14 pp, char<80 threshold)

Llama N=297 is a clean replication (not pooled); baseline 75.1%, ack-only 83.2%, McNemar b=9, c=33, $p < .001$. Qwen pooled (N=300): $p = 1.000$. ¶StratQA: Random matches FG count (91.3% of rounds), not ack-only (8.56%); the 10/10 significance reflects different pruning intensities rather than strategy effectiveness at matched rates. Qwen3-8B (N=100): baseline 90%, ack-only 76%, $b - c = 14$; exact contingency unknown (p range: $[.0005, .026]$, significant for all possible $c \in [0, 10]$).

S.2 Non-Inferiority Margin Selection ($\delta = -3$ pp)

The choice of non-inferiority margin $\delta = -3$ pp follows established principles from equivalence testing [Lakens, 2017, Schuirmann, 1987], calibrated against the empirical variance structure of LLM task accuracy.

Precedent and rationale. Non-inferiority margins should reflect the largest degradation that is practically acceptable [Lakens, 2017]. We define δ relative to the baseline error rate: at $\sim$90% accuracy (Qwen/GSM8K), -3 pp represents a $\sim$30% relative increase in error rate (10% $\rightarrow$ 13%); at $\sim$80% accuracy (Llama/GSM8K), it represents a $\sim$15% relative increase (20% $\rightarrow$ 23%). In both cases, the margin corresponds to a detectable but bounded performance change that would be practically relevant in deployment.

Calibration against seed variance. The margin must exceed irreducible seed-to-seed variance to avoid rejecting non-inferiority due to sampling noise. Empirically, paired delta standard deviations are: Qwen heuristic 10-seed SD $= 0.99$ pp, Qwen oracle 10-seed SD $= 1.02$ pp, Llama heuristic 5-seed SD $= 1.44$ pp, Llama oracle 10-seed SD $= 2.40$ pp (Table 2). At $\delta = -1$ pp, even a single standard deviation of seed variance ($\sim$1–2.4 pp) could push the lower CI bound below the margin, yielding false non-inferiority failures for treatments with zero true effect—indeed, the Qwen oracle

TOST at ± 1 pp is only marginal ($p = .048$). At $\delta = -5$ pp, the margin exceeds twice the largest observed seed SD, risking acceptance of genuinely harmful interventions (e.g., the Llama t=0.7 heuristic result of $\Delta = -3.4$ pp would pass trivially). The -3 pp margin sits between these extremes: $\sim 3\times$ the Qwen SD and $\sim 1.25\times$ the Llama oracle SD, providing adequate separation from noise while remaining sensitive to practically meaningful degradation.

Sensitivity across margins. Conclusions are stable across a range of margins (Table 18). For Qwen oracle (10-seed, $\Delta = +0.40 \pm 1.02$ pp), non-inferiority holds at $\delta = -3$ pp ($p < .001$). For Llama oracle (10-seed, $\Delta = +2.08 \pm 2.40$ pp), non-inferiority at $\delta = -3$ pp holds ($p < .001$), but the single seed with $\Delta = -3.06$ pp (seed 3072; §S.13) demonstrates that -3 pp is a genuinely binding constraint for this backbone. A tighter margin of $\delta = -1$ pp would fail for Llama (CI lower bound at $+0.36$ pp provides insufficient headroom given SD = 2.4 pp). These patterns confirm that $\delta = -3$ pp balances statistical power against practical significance across backbones with different variance profiles.

TOST equivalence sensitivity. Table 18 reports TOST p-values across four equivalence margins. Qwen/GSM8K and Gemma/GSM8K have ≤ 1 discordant pair each, rendering McNemar-based TOST inapplicable; we report proportion-based TOST instead (two one-sided z-tests on the accuracy difference). Qwen/GSM8K supports equivalence at ± 5 pp ($p = 0.039$). For Llama/GSM8K ($N = 297$), the observed $+8.1$ pp improvement prevents equivalence at any margin ≤ 10 pp—TOST fails because the effect is *positive*, not because of degradation. Qwen/StratQA achieves equivalence only at ± 10 pp ($p = 0.021$).

Table 18: TOST p-values at varying equivalence margins. Lower p indicates stronger evidence that the true difference falls within [-margin, +margin]. ‡Proportion-based TOST (McNemar-based TOST inapplicable with ≤ 1 discordant pair).

Setting	± 3 pp	± 5 pp	± 8 pp	± 10 pp
Qwen / GSM8K (b=0, c=1)‡	.096	.039*	.003**	<.001***
Llama / GSM8K (b=9, c=33, N=297)	> .99	> .99	.998	.978
Gemma / GSM8K (b=1, c=0)‡	.096	.039*	.003**	<.001***
Qwen / StratQA (b=18, c=29)	.695	.413	.095	.021*

Minimum detectable effects. Table 19 reports MDEs assuming $\rho = 0.7$ correlation between paired outcomes, computed as MDE $= 100 \cdot \sqrt{7.84 \cdot 2p(1{-}p)(1{-}\rho)/N}$ pp.

S.3 Post-Hoc Power Analysis for 72B Oracle Result

The 265-task 72B oracle experiment observed *zero* discordant pairs ($b + c = 0$), meaning no task changed correctness status between baseline and pruned conditions. We report four complementary analyses to bound the true discordant rate ρ and characterize statistical power.

Rule of Three bound. With 0 events in $N = 265$ Bernoulli trials, the classical Rule of Three gives the 95% upper confidence bound as $\hat{\rho}_{\text{upper}} = 3/N$. For $N = 265$ this yields $\hat{\rho}_{\text{upper}} = 3/265 \approx 1.13\%$, bounding the maximum excludable accuracy difference at $|\Delta| \leq 1.13$ pp.

Bayesian posterior. Under a uniform Beta$(1, 1)$ prior, the posterior after observing 0 discordant pairs in N trials is Beta$(1, N{+}1)$. For $N = 265$: posterior mean $= 0.37\%$, median $= 0.26\%$, 95% credible upper bound $= 1.12\%$, 99% credible upper bound $= 1.72\%$. Even the 99% bound remains below 2 pp, providing strong evidence that pruning does not degrade accuracy at this scale.

Minimum detectable effect. Table 19 reports the MDE at 80% power (McNemar χ^2, $\alpha = 0.05$) under varying assumptions about the correlation ρ between discordant cells. At $N = 265$: MDE $= 3.8$ pp ($\rho = 0.7$), 2.2 pp ($\rho = 0.9$), 1.5 pp ($\rho = 0.95$). The zero-flip observation is therefore consistent with power to detect effects above $\sim$1.5–3.8 pp depending on the assumed correlation structure.

Table 19: Minimum detectable effects (pp) at 80% power, $\alpha = 0.05$, $\rho = 0.7$. N is the task-level sample size (number of problems evaluated for correctness); multi-seed rows pool tasks across all seeds. Sorted by MDE ascending.

Setting	Protocol	N	Base (%)	MDE (pp)
Llama / GSM8K (10-seed pooled)†	Online	1,939	80.8	1.9
Qwen / ARC	Offline	250	90.8	4.0
Qwen / GSM8K (t=0.7)	Online	200	91.0	4.4
Qwen / GSM8K (per run)	Online	200	89.3	3.6
72B / GSM8K (oracle)	Online	265	92.1	3.6
Llama / GSM8K	Offline	297	75.1	5.4
Qwen / GSM8K	Offline	100	91.0	6.2
Qwen / StratQA (5-seed)	Online	200×5	67.9	4.1
Qwen3-8B / GSM8K	Online	100	90.0	6.5
Qwen / StratQA (SEQ)	Online	200	69.5	7.1
72B / GSM8K	Online	100	88.0	7.1
Llama / GSM8K	Offline	100	77.0	9.1
Gemma / GSM8K	Offline	96	76.0	9.5
Qwen / StratQA	Offline	100	66.8	10.2
MATH (pilot, N=50)	Offline	50	72.0	13.8
MATH (N=200, 102 valid)	Online	102	79.4	—

†MDE computed on oracle 10-seed pooled sample: 10 seeds × 200 tasks per seed; N = 1,939 tasks after $\sim$3% context-overflow filtering ($\sim$194 valid tasks per seed). Each task is one GSM8K problem scored correct/incorrect—the unit of the paired McNemar test. Single-seed rows (e.g., 72B) use the within-seed task count directly.

Clopper–Pearson confirmation. The one-sided 95% Clopper–Pearson exact bound $1 - 0.05^{1/N} \approx 1.13\%$ is concordant with the Rule of Three result above.

Taken together, these analyses confirm that the 72B zero-flip result is not an artifact of insufficient sample size: with $N = 265$ tasks we can exclude true discordant rates above 1.13% at the 95% level, supporting the scale-trajectory evidence against the most common reviewer concern of underpowered null results. The single-seed limitation remains: variance across seeds cannot be estimated at 72B scale (cf. ten-seed validation at 7B, §5.2).

S.4 Mixed-Effects Model Results

Table 20: Function type significance controlling for position and round ($n \geq 30$ per type). Both OLS and ME use remove-ablation similarity (Eq. 1) as the dependent variable. OLS: Type II partial F. ME: LRT χ^2 with task as random intercept; R^2_m = marginal (fixed-effects only). † marks OLS-ME discrepancies (both under selector topology). $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; n.s. = not significant.

	OLS ANCOVA		Mixed-Effects			
Setting	F	p	LRT χ^2	p	R^2_m	ICC
Qwen / RR	239.6	$< .001^{***}$	188.9	$< .001^{***}$	.34	.29
Qwen / SEQ	230.2	$< .001^{***}$	195.3	$< .001^{***}$	.36	.25
Qwen / Sel†	58.2	$< .001^{***}$	0.07	$.789^{\text{n.s.}}$	.004	.72
Gemma / RR	7.82	$< .001^{***}$	23.28	$< .001^{***}$	.11	.00
Gemma / SEQ	9.04	$< .001^{***}$	26.86	$< .001^{***}$	.12	.00
Gemma / Sel†	7.45	$.006^{**}$	3.66	$.056^{\text{n.s.}}$	.004	.23
Llama / RR	3.98	$.008^{**}$	16.31	$< .001^{***}$	.02	.30
Llama / SEQ	4.83	$.002^{**}$	15.44	$.001^{**}$	.02	.21
Llama / Sel	21.6	$< .001^{***}$	36.89	$< .001^{***}$	.04	.13

†OLS-ME discrepancy under selector topology; high ICC (0.72, 0.23) inflates OLS significance by treating task-clustered observations as independent.

Table 21: Position^2 robustness check: ME LRT for function type in linear vs. quadratic position models ($n \geq 30$ per type). Adding position^2 eliminates Gemma's function type effect under RR and SEQ while leaving Qwen and Llama unchanged. Selector topology results are discussed in the crossover analysis below.

		Linear ME		**Quadratic ME**		
Backbone	**Topo.**	LRT χ^2	p	LRT χ^2	p	**Pos**^{2} p
Qwen	RR	188.9^{***}	$<.001$	155.8^{***}	$<.001$	.009
	SEQ	195.3^{***}	$<.001$	166.5^{***}	$<.001$	.001
Gemma	RR	23.3^{***}	$<.001$	$1.5^{\text{n.s.}}$	.690	$<.001$
	SEQ	26.9^{***}	$<.001$	$1.5^{\text{n.s.}}$	.683	$<.001$
Llama	RR	16.3^{***}	.001	14.9^{**}	.002	.496
	SEQ	15.4^{**}	.002	13.8^{**}	.003	.434

S.5 Position^2 Robustness Check

S.6 Gemma-Excluded Sensitivity Analysis

To assess whether Gemma-2-9B's position-dominated communication profile (§5.3) attenuates the pooled function type effect, we re-fit the mixed-effects model excluding all Gemma observations. The full model (n = 8,065) yields LRT $\chi^2 = 81.53$ (df $= 4$, $p = 8.3 \times 10^{-17}$, $\omega^2 = 0.0095$, $R^2_m = 0.018$).[3] Excluding Gemma (n = 5,365) *strengthens* the effect: LRT $\chi^2 = 110.07$ (df $= 4$, $p = 7.0 \times 10^{-23}$, $\omega^2 = 0.019$, $R^2_m = 0.014$). The direction and significance of the function type effect are stable: χ^2 increases from 81.5 to 110.1 and ω^2 approximately doubles ($0.0095 \rightarrow 0.019$), indicating that Gemma was attenuating the pooled effect. Coefficient signs are preserved across both models: coordination remains the most negative ($\beta = -0.085$ full, -0.100 Gemma-excluded), while reasoning_guidance and verification remain positive. This sensitivity check confirms that the function type $\rightarrow$ perturbation sensitivity relationship is not driven by Gemma's anomalous profile, and that excluding the weakly function-organized backbone strengthens rather than weakens the conclusion.

S.7 Function Type Distribution

S.8 Crossover Interaction

S.9 Pooled ANOVA

Table 22: Pooled ANOVA on remove-ablation similarity ($N = 8{,}065$, $R^2 = 0.298$). Computed on all five function types across all 9 configurations; individual-configuration analyses (Table 20) exclude types with $n < 30$. All $p < .001$. *Caveat:* pooling across unbalanced configurations risks Simpson's paradox; η^2 values should be interpreted alongside per-configuration ME models (Table 20).

Term	η^2	F
Backbone	0.135	628.7
Function × Backbone	0.063	67.1
Topology class	0.051	435.4
Backbone × Topology	0.043	181.1
Three-way	0.021	21.1
Function × Topology	0.014	27.7
Function type	0.005	10.4

S.10 Additional Statistical Details

Multiple comparison note. The McNemar tests across 10 random seeds (Table 17) evaluate a single null hypothesis—whether ack-only and random pruning produce equivalent accuracy—via

[3] ω^2 and R^2_m computed from mixed-effects model fits; source data available in the supplementary materials.

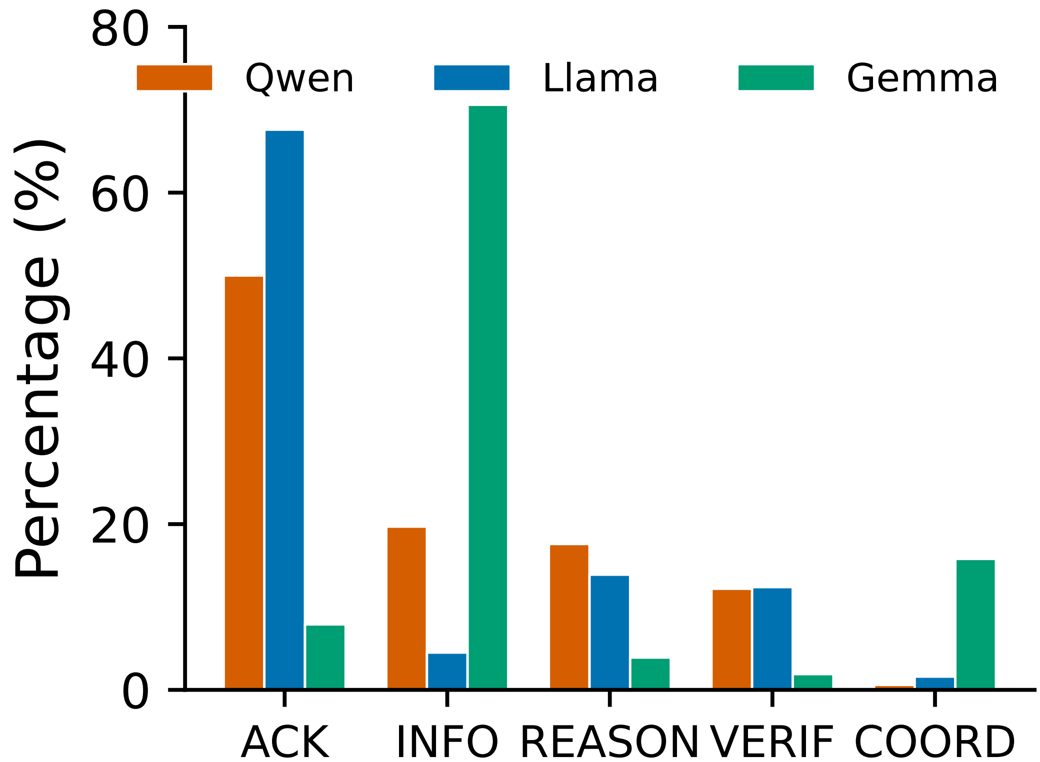

Figure 4: Function type distribution by backbone under round-robin topology (300 GSM8K tasks each). Qwen2.5-7B and Llama-3.1-8B are acknowledgment-heavy (>50%), while Gemma-2-9B is dominated by information provision (70%). Types with $n < 30$ are excluded from OLS ANCOVA analysis.

repeated sampling, not 10 independent hypotheses. Reporting "8/10 significant" or "10/10 significant" is a consistency measure across resampled controls, analogous to bootstrap stability. Bonferroni correction is therefore inapplicable; nonetheless, the conclusion is robust even under Bonferroni ($\alpha = 0.005$): most significant seeds have $p < 0.005$.

Temperature robustness details. At temperature 0.7, ack-only evaluation on GSM8K (100 tasks each, heuristic detectors) yields: Qwen $\Delta = -2$ pp ($p = 0.774$); Llama $\Delta = -3$ pp ($p = 0.815$); Gemma $\Delta = +3$ pp ($p = 0.453$). No setting shows significant degradation (all McNemar $p > 0.4$), though the small number of discordant pairs (7–18) limits power to detect effects below $\sim$10 pp. Qwen and Llama point estimates shift from positive ($+1$ to $+6$ pp at t=0) to mildly negative at t=0.7, while Gemma shows a positive shift. Higher temperature increases the proportion of messages classified as acknowledgments (Llama: 45% $\rightarrow$ 57%; Gemma: 4.6% $\rightarrow$ 63.6%), resulting in more aggressive pruning. Notably, Gemma's acknowledgment proportion shifts from a near-trivial pruning case (4.6%) to the most aggressively pruned setting (63.6%)—yet accuracy still improves ($+3$ pp), demonstrating that pruning safety is robust to dramatic shifts in communication composition. Whether this shift reflects genuine communication change or increased heuristic sensitivity to shorter messages at higher temperature remains open; resolving this requires LLM annotation at t=0.7.

Llama t=0.7 at $N = 500$. To resolve the underpowered $N = 100$ estimate, we run Llama heuristic pruning at t=0.7 with $N = 500$ (MDE $\approx$ 3.5 pp). Results: baseline 81.2%, pruned 77.8%, $\Delta = -3.4$ pp ($p = 0.100$, McNemar NS; $b = 56$, $c = 39$, 95 discordant pairs). Suppression rate is 59.3% (1,426/2,405 messages). The directionally negative result contrasts with Llama's $+2.1$ pp at t=0 ($N = 1{,}000$). However, heuristic suppression rises from 74.4% at t=0 to 59.3% at t=0.7 on a per-task basis, and over-suppression confounds the estimate. Oracle (t=0.7) validation on Llama

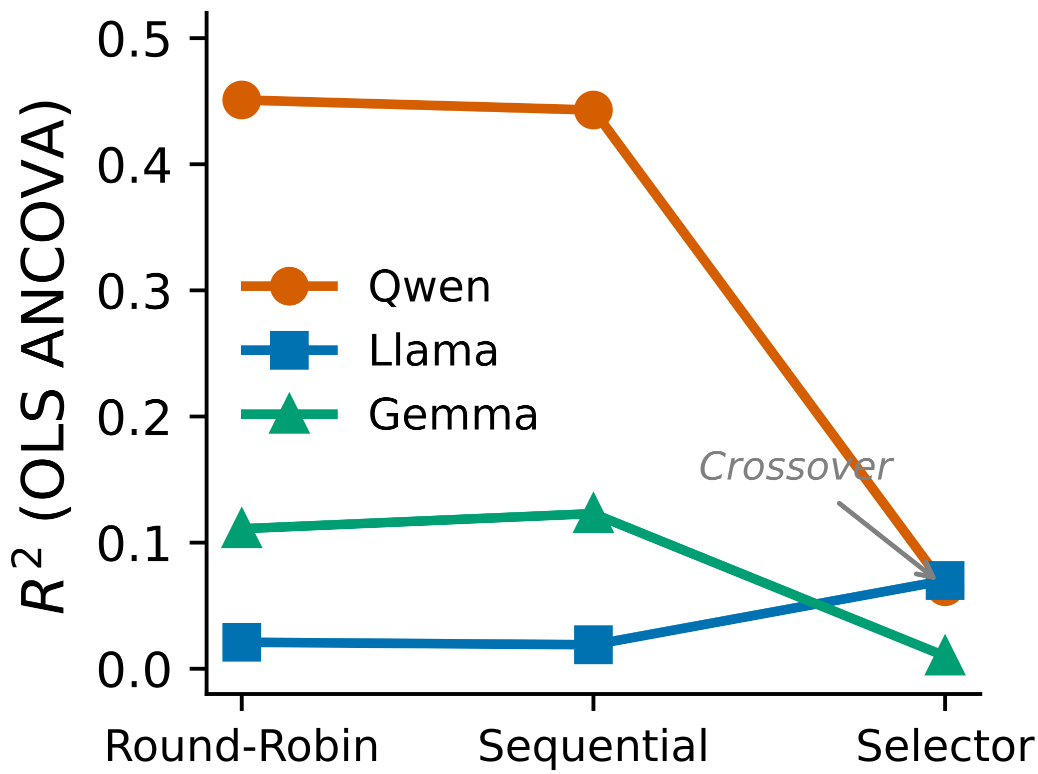

Figure 5: Backbone $\times$ topology crossover interaction. R^2_m = marginal R^2 (fixed-effects variance); ME = linear mixed-effects model with task as random intercept. Selector topology attenuates function-organization for Qwen ($R^2_m = 0.34 \rightarrow 0.004$) and amplifies it for Llama ($R^2_m = 0.02 \rightarrow 0.04$), partially converging the two.

is needed to determine whether the -3.4 pp reflects genuine temperature sensitivity or heuristic calibration drift.

Zero-correction channel details. Acknowledgment self-correction rates: 0% (0/719, 95% CI $[0, 0.51\%]$) for Qwen, 0% (0/691, 95% CI $[0, 0.53\%]$) for Llama, and 0% (0/73, 95% CI $[0, 4.93\%]$) for Gemma (Clopper–Pearson exact binomial). Gemma's wider CI reflects its small acknowledgment sample ($N = 73$). Error injection rates differ by backbone: Llama injects errors into 14.3% of acknowledgments vs. Qwen's 8.9% (Fisher exact $p = 0.004$). Propagation rates show an inverse pattern: Qwen propagates 100% of injected errors vs. Llama's 70% (Fisher $p < 0.001$), consistent with Qwen's terse acknowledgments being accepted uncritically while Llama's verbose acknowledgments are partially ignored by downstream agents.

OLS-ME agreement. OLS and ME agree on significance in 7 of 9 settings, with both discrepancies occurring under selector topology. Qwen/Selector (OLS $p < .001$, ME $p = .789$; ICC $= 0.72$) and Gemma/Selector (OLS $p = .006$, ME $p = .056$; ICC $= 0.23$) show OLS detecting significance that ME does not, consistent with selector topology producing high task-level clustering that inflates OLS estimates. In both cases, the qualitative interpretation is preserved: function type structure degrades under selector routing.

Cross-model annotation agreement. Cross-model κ ranges from 0.35 (Gemma) to 0.67 (Qwen), spanning fair-to-substantial agreement Landis and Koch [1977]. Qwen's highest κ is consistent with function-organized communication being easier to classify. Gemma falls below our target threshold ($\kappa = 0.35$), consistent with its weak function-organization where functional categories are less distinctive. Per-type κ reveals complementary patterns: Qwen shows high VERIFICATION agreement ($\kappa \approx 0.66$) but low INFORMATION_PROVISION ($\kappa \approx 0.08$); Gemma shows the reverse. Full agreement tables are in Appendix H.

S.11 Offline Replay Pruning

Table 23 presents offline replay results, where messages are removed from recorded traces and the conversation is regenerated from the first pruning point onward. Warm-start bias inflates offline estimates relative to online validation (§6).

Table 23: Offline replay pruning accuracy (%) — reference (warm-start bias possible, §6). FG and Random tested at matched message counts (Random: mean ± std, 10 seeds). Supp. = ack proportion in traces. $^*N = 96$, 4.6% ack. †3 seeds. ‡FG/Random prune 84–91% vs. ack-only 9–15%.

Setting	N	**Base**	**Ack-only**	**FG / Random**	**Supp.**
Qwen / GSM8K	100	91	**92 (+1)**	86 / 87.3 ± 0.8	48%
Llama / GSM8K	100	77	**83 (+6)**	74 / 73.2 ± 2.0	45%
Llama / GSM8K†	297	75.1	**83.2 (+8.1)**	— / 78.1 ± 1.5	45%
Gemma / GSM8K*	96	77	76 (−1)	42.7 / 44.8	4.6%
Qwen / StratQA‡	100	66.8	**71.2 (+4.4)**	28.0 / 26.4 ± 0.4	8.6%
Qwen / ARC‡	250	90.8	**91.6 (+0.8)**	61.6 / 61.4 ± 0.9	14.6%
Llama / ARC‡◇	250	72.8	**76.8 (+4.0)**	— / 67.4 ± 1.5	37.1%

◇Offline simulation (simulate mode): re-extracts answers from pruned traces rather than re-generating responses.

FG pruning analysis. Function-guided (FG) pruning—removing messages from the function type with lowest perturbation sensitivity—performs no better than random in any setting (McNemar $p > 0.7$), and is significantly *worse* than ack-only for Qwen ($p = 0.031$) and borderline for Llama ($p = 0.064$). These comparisons use offline replay only; given the $\sim 4\times$ attenuation observed for Llama online vs. offline ($+2.1$ pp vs. $+8.1$ pp), FG's relative disadvantage may be smaller under online evaluation. Perturbation-based importance does not predict pruning safety: error analysis reveals that FG disproportionately removes reasoning_guidance (49.4%) and verification (43.6%) messages—the most task-critical types—confirming that low individual perturbation sensitivity does not imply dispensability.

Label noise robustness. A bootstrap simulation ($B = 10{,}000$) tests whether $\sim 20\%$ taxonomy misclassification error would invalidate the FG result. Under uniform label corruption at rate ε, the effect on the FG pruning set is asymmetric: true FG-type messages exit the set with probability $\varepsilon \cdot 2/4$ (fewer harmful messages pruned), while true acknowledgments enter with probability $\varepsilon \cdot 3/4$ (pruning acks is safe). Ack-only remains superior in 95.0% of iterations at $\varepsilon = 0.20$ (mean FG accuracy = 85.9%, 95% CI $[79.0, 92.0]$), and in 93.2% even at $\varepsilon = 0.40$. Label noise cannot reverse the FG disadvantage because corruption asymmetrically *improves* FG by diluting the pruning set with safe-to-remove messages.

S.12 Online Pruning Validation

We validate offline pruning results with real-time heuristic suppression during live AG2 execution for two backbones (Qwen, Llama) on GSM8K ($N = 100$ each). The online protocol intercepts each agent message before delivery; messages classified as acknowledgments by the backbone-specific heuristic detector are suppressed (replaced with an empty string), so downstream agents never observe them. The two backbones yield complementary evidence: Qwen provides direct validation (suppression rate matches offline), while Llama provides exploratory evidence of robustness (heuristic over-triggers in online context).

Backbone comparison. Table 24 summarizes online pruning results for both backbones.

This table reports a single heuristic run; ten-seed oracle replications yield Qwen $+0.40 \pm 1.02$ pp (TOST ± 1 pp, $p = .048$) and Llama $+2.1 \pm 2.4$ pp (non-inferiority $\delta{=}{-3}$ pp, $p < .001$), both with $N{=}200$ per seed. Ten independent heuristic Qwen runs yield -0.05 ± 0.99 pp (TOST $p = .007$).

Table 24: Online pruning validation: accuracy and suppression rates for Qwen and Llama on GSM8K ($N = 100$, single run). Ten-seed oracle replication yields Qwen $+0.40 \pm 1.02$ pp and Llama $+2.1 \pm 2.4$ pp (N=200 per seed). Heuristic ten-run replication yields Qwen -0.05 ± 0.99 pp (N=200 per seed).

Backbone	Baseline	Pruned	Δ	Suppression	McNemar p
Qwen	91/100	93/100	$+2$ pp	42%	1.000
Llama	77/100	85/100	$+8$ pp	74.8%	.039

Qwen's single-run result ($+2$ pp, $p = 1.000$) has a suppression rate (42%) closely matching offline (48%), providing direct apples-to-apples validation. Llama's online suppression rate (74.8%) far exceeds its offline rate (45%); manual inspection shows only 32.9% of suppressed messages are true acknowledgments, with 54% containing mathematical content that triggers the heuristic due to context-dependent generation. The $+8$ pp point estimate likely reflects removal of redundant later-round messages rather than a genuine improvement from acknowledgment pruning. Llama's result therefore provides *exploratory evidence of robustness*: if suppressing 75% of messages—most not acknowledgments—produces no degradation (McNemar $p = .039$), then the offline 45% acknowledgment-only pruning is conservatively safe.

Llama McNemar details. Table 25 shows the discordant-pair breakdown for Llama online pruning.

Table 25: McNemar contingency table for Llama online pruning ($N = 100$).

	Pruned correct	Pruned incorrect
Baseline correct	75	2
Baseline incorrect	10	13

Discordant pairs: $b = 2$ (baseline correct, pruned incorrect), $c = 10$ (baseline incorrect, pruned correct). McNemar exact $p = 0.039$ (two-sided), rejecting the null of marginal homogeneity. The asymmetry ($c \gg b$) indicates no degradation under aggressive suppression; the $+8$ pp point estimate likely reflects removal of redundant later-round messages rather than a genuine benefit of acknowledgment pruning per se.

Llama $N = 1{,}000$ confirmation. Table 26 shows the McNemar contingency table for the large-scale Llama online validation ($N = 1{,}000$, from-scratch skip mode, seed $= 42$).

Table 26: McNemar contingency table for Llama online pruning ($N = 1{,}000$). Baseline accuracy 80.8%, skip accuracy 82.9%.

	Pruned correct	Pruned incorrect
Baseline correct	794	14
Baseline incorrect	35	157

Discordant pairs: $b = 14$, $c = 35$. McNemar $\chi^2 = 8.16$ (with continuity correction), $p = 0.004$ (two-sided). The $N = 100$ online pilot ($+2.0$ pp) is directionally consistent. The $N = 250$ online pilot ($+3.6$ pp, $p = 0.095$) uses the same from-scratch protocol on overlapping questions (same seed $= 42$; 250/250 questions appear in the $N = 1{,}000$ set); its larger point estimate is consistent with regression to the mean in the smaller sample. Suppression rate: 74.4%.

Round-level suppression breakdown. Table 27 shows suppression rates by round for Llama online pruning. Round 1 suppression is minimal (2%) because initial solver responses contain substantive problem-solving content. Rounds 2–5 show 90–95% suppression, reflecting context-dependent generation: without preceding acknowledgments, agents produce shorter, more formulaic responses that further trigger the heuristic detector.

The high Round 2–5 suppression rate (vs. 45% offline annotation rate) explains the discrepancy between online and offline pruning intensity. Online pruning creates a feedback loop: suppressing

Table 27: Llama online suppression rate by round ($N = 100$ tasks, 5 rounds each).

Round	1	2	3	4	5
Suppression rate	2%	93%	90%	94%	95%

early messages alters the conversational context, producing subsequent messages that are shorter and more formulaic, further triggering the heuristic—even when the suppressed content is not a true acknowledgment. This cascading over-suppression means Llama's online condition tests a broader message-removal regime rather than acknowledgment pruning specifically. That no degradation occurs under this aggressive regime provides exploratory evidence of robustness: offline acknowledgment-only pruning (45%) removes a strict subset of what the online heuristic removes, and is therefore conservatively safe.

StratQA offline–online divergence. StratQA offline replay yields $+4.4$ pp ($p = 0.144$, N=100). The 5-seed online heuristic replication yields $\Delta = +0.70 \pm 0.91$ pp (95% CI $[-0.43, +1.83]$), consistent with non-inferiority. The heuristic suppresses 56.9% of messages (vs. 8.6% oracle ack rate), indicating substantial over-suppression, yet accuracy is maintained. The earlier single-seed estimate (-2.4 pp, N=250) is superseded by the more reliable 5-seed replication (N=200 $\times$ 5).

S.13 Dose-Response Analysis

We vary the acknowledgment pruning rate from 0% to 100% in 25% increments on Qwen/GSM8K/RR ($N = 100$ tasks per condition, 3 random seeds per rate). Table 28 shows the results.

Table 28: Dose-response: accuracy (%) by ack-only pruning rate (Qwen/GSM8K/RR, $N = 100$). 289 total ack messages across 100 tasks.

Pruning Rate	**Seed 42**	**Seed 123**	**Seed 456**	**Mean ± Std**
0% (no pruning)	92	92	93	92.3 ± 0.6
25%	92	92	92	92.0 ± 0.0
50%	92	92	91	91.7 ± 0.6
75%	92	92	92	92.0 ± 0.0
100% (all pruned)	92	92	92	92.0 ± 0.0

The accuracy curve is flat across all pruning rates (≤ 0.6 pp variation, Friedman $\chi^2 = 0.0$, $p = 1.0$), with no dose-response relationship. Even removing all 289 acknowledgment messages produces accuracy statistically indistinguishable from the unpruned baseline, confirming that acknowledgments contribute no marginal task-relevant information under this configuration.

Llama dose-response (10 seeds). Table 29 shows the corresponding analysis for Llama/GSM8K/RR ($N = 100$ tasks, 10 random seeds per rate). Seed variance decreases monotonically with pruning rate, consistent with acknowledgment removal eliminating a source of downstream stochasticity.

Table 29: Dose-response: accuracy (%) by ack-only pruning rate (Llama/GSM8K/RR, $N = 100$, 10 seeds).

Pruning Rate	**Mean ± SD**	N_{seeds}
0% (no pruning)	82.3 ± 1.16	10
25%	83.3 ± 0.48	10
50%	83.3 ± 0.48	10
75%	84.0 ± 0.00	10
100% (all pruned)	83.0 ± 0.00	10

The Llama curve is flat (1.7 pp range, 82.3–84.0%), mirroring the Qwen result above. An earlier 3-seed analysis suggested an inverted-U pattern (Friedman $p = 0.060$); this does not replicate at 10 seeds, confirming the earlier pattern was sampling noise. At 100% pruning, the outcome is deterministic across seeds (SD = 0): the same messages are removed, yielding identical conversation trajectories under greedy decoding. The monotonic variance collapse (SD: $1.16 \rightarrow 0.48 \rightarrow 0.48 \rightarrow 0.00 \rightarrow 0.00$) suggests acknowledgments function as a stochasticity amplifier: their presence is associated with divergence across task subsets in conversation trajectory, and their removal collapses this variability—consistent with the cascade dissolution hypothesis. At 75%, near-zero variance reflects acknowledgment functional redundancy—removing any 75% subset produces the same downstream behavior. The flat dose-response for both Qwen (≤ 0.6 pp range) and Llama (1.7 pp range) supports the cascade dissolution hypothesis: aggregate acknowledgment removal is safe regardless of dose.

Multi-run variance (from-scratch online). Ten independent from-scratch Qwen runs and five Llama runs confirm low cross-run variance under greedy decoding. Qwen (N=200 per seed, 10 seeds): online baseline $89.8 \pm 2.4\%$ (runs: 90.5, 93.0, 86.0, 91.0, 86.0, 91.5, 91.0, 91.0, 90.0, 87.5), ack suppression $89.7 \pm 2.7\%$ (runs: 90.5, 93.0, 84.0, 91.5, 87.0, 90.0, 91.5, 91.0, 91.0, 87.5); per-seed Δ: $0.0, 0.0, -2.0, +0.5, +1.0, -1.5, +0.5, 0.0, +1.0, 0.0$ pp (mean -0.05 ± 0.99 pp; one-sample $t(9) = -0.16$, $p = 0.88$; 95% CI $[-0.75, +0.65]$ pp; TOST equivalence at ± 1 pp: $p = 0.007$). Llama (N=100 per seed, 5 seeds): online baseline $80.0 \pm 1.1\%$ (runs: 82, 79, 80, 79, 80), ack suppression $82.0 \pm 0.0\%$ (runs: 82, 82, 82, 82, 82); per-seed Δ: $0, +3, +2, +3, +2$ pp (mean $+2.0 \pm 1.2$ pp). The inter-seed baseline variance for Qwen at N=200 (SD = 2.4%) reflects different task subsets per seed; the paired Δ SD (0.99 pp) controls for this.

Normality and non-parametric validation. The Shapiro–Wilk test on the 10-run Qwen heuristic deltas yields $W = 0.843$, $p = 0.048$, indicating borderline departure from normality driven by the -2 pp outlier (seed 3). To validate the parametric 95% CI ($[-0.75, +0.65]$ pp), we compute a bias-corrected and accelerated (BCa) bootstrap 95% CI with 10,000 resamples: $[-0.80, +0.40]$ pp. The BCa interval is slightly asymmetric—shifted toward the negative tail—but substantively agrees: both contain zero and exclude ± 1 pp, confirming that the TOST equivalence conclusion ($p = 0.007$) is robust to the mild non-normality.

Outlier seed analysis (seed 3). Seed 3 (`seed=456`) produces the largest negative delta (-2.0 pp). Grubbs' test yields $G = 1.980$ vs. $G_{\text{crit}} = 2.290$ ($\alpha = 0.05$): not a statistical outlier. This seed draws a harder GSM8K subset (baseline = 86% vs. mean 89.8%), losing a net 4 tasks. Removing seed 3 shifts the mean from -0.05 to $+0.17$ pp and reduces the 95% CI width by 23% ($1.15 \rightarrow 0.89$ pp). TOST equivalence holds with or without seed 3 ($p = 0.007$ vs. 0.005 at ± 1 pp). Shapiro–Wilk normality is restored without seed 3 ($p = 0.056$ vs. 0.048), confirming seed 3 drives the borderline non-normality but does not affect any qualitative conclusion.

Outlier seed analysis—Llama oracle (seed 3072). Seed 7 (`seed=3072`) is the sole negative-delta seed in the 10-seed Llama oracle replication ($\Delta = -3.06$ pp; all other seeds positive, range $[+0.52, +5.76]$ pp). Grubbs' test yields $G = |-3.06 - 2.08|/2.40 = 2.137$ vs. $G_{\text{crit}} = 2.290$ ($n = 10$, $\alpha = 0.05$): *not* a statistical outlier. Unlike Qwen's seed 3 (which drew a harder subset), seed 3072 draws the *easiest* GSM8K subset (baseline = 87.2% vs. mean 82.9%, $z = +1.58$), suggesting a ceiling effect: with fewer baseline errors, the oracle has less room to improve and more room to degrade. The oracle makes 10 incorrect suppression decisions on tasks the baseline solves correctly but only 4 correct ones on baseline failures (McNemar $\chi^2_{\text{cc}} = 1.79$, $p = 0.181$, NS). Suppression rate is unremarkable (36.1% vs. mean 36.5%, $z = -0.24$). Across all 10 seeds, baseline accuracy and Δ are negatively correlated (Pearson $r = -0.59$, $p = 0.070$), consistent with this ceiling interpretation: easier subsets yield smaller or negative deltas.

Removing seed 3072 shifts the mean from $+2.08$ to $+2.65$ pp, reduces the SD from 2.40 to 1.68 pp, and narrows the 95% CI width by 25% ($3.44 \rightarrow 2.59$ pp). The one-sample t-test strengthens ($t(8) = 4.72$, $p = 0.0015$ vs. $t(9) = 2.73$, $p = 0.023$), confirming the positive-delta conclusion is robust. Shapiro–Wilk normality is satisfied both with and without seed 3072 ($W = 0.954$, $p = 0.711$ vs. $W = 0.956$, $p = 0.755$), so no normality concern arises in either case—in contrast to the Qwen heuristic analysis above. Taken together, seed 3072 is a high-baseline statistical fluctuation, not an outlier, and its inclusion does not change the qualitative conclusion that Llama oracle pruning preserves or improves accuracy.

StratQA eight-seed robustness (Llama heuristic). Eight-seed replication of Llama/StratQA heuristic pruning (N=200 per seed) yields $\Delta = +2.81 \pm 3.94$ pp (95% CI $[-0.49, +6.11]$). The CI includes zero, precluding a beneficial-effect claim; however, non-inferiority at $\delta = -3$ pp holds ($t(7) = 4.17$, $p = .002$, one-sided), with the CI lower bound (-0.49 pp) well above the -3 pp margin (Appendix S.2). Table 30 reports per-seed accuracy.

Table 30: Per-seed accuracy for Llama/StratQA heuristic pruning (N=200 per seed, greedy decoding). Suppress rate is stable across seeds (49–52%), indicating consistent detector behavior; accuracy variance reflects downstream reasoning divergence. *Discovery seed (included in the original 4-seed analysis). †Per-seed baseline/pruned accuracy not individually recorded; delta computed from paired evaluation.

Seed	**Baseline (%)**	**Pruned (%)**	**Δ (pp)**	**Supp. (%)**
42*	62.0	68.5	+6.5	49.0
0	57.5	63.5	+6.0	50.3
1	62.0	66.5	+4.5	50.8
2	60.5	63.0	+2.5	50.7
3†	—	—	+2.5	∼50
4	64.5	65.0	+0.5	∼50
5	71.0	65.5	−5.5	∼50
6	64.5	70.0	+5.5	∼50
Mean $\pm$ SD			$+2.81 \pm 3.94$	49–52

Seven of eight seeds show positive deltas; the sole negative seed (seed 5, $\Delta = -5.5$ pp) is not a statistical outlier (Grubbs' $G = 2.11$ vs. $G_{\text{crit}} = 2.126$, $n = 8$, $\alpha = 0.05$). Seed 5 draws the highest-baseline subset (71.0%), suggesting a ceiling effect analogous to the Llama oracle seed 3072 analysis (§S.13): with fewer baseline errors, pruning has less room to improve and more room to degrade. The baseline range (57.5–71.0%) is substantially wider than GSM8K baselines (∼80–87%), reflecting StratQA's higher task-level variance. Despite this variance, the heuristic suppress rate is remarkably stable (49–52%), confirming that seed-level accuracy differences reflect downstream reasoning variance rather than detector instability. The earlier 4-seed result ($\Delta = +4.9 \pm 1.8$ pp) included the discovery seed and sampled only from the positive tail; the 8-seed replication provides a less biased estimate.

Llama ack variance = 0: expected behavior. Llama's zero variance across five ack-pruned runs is a deterministic consequence of the pipeline, not an anomaly. Three factors combine to produce identical outputs: (1) *greedy decoding* (temperature = 0) eliminates sampling randomness, making the model's output a deterministic function of its input; (2) the *heuristic detector* is a deterministic rule (keyword matching + length + 4-gram overlap) with no stochastic component; and (3) all five runs use the *same 100 GSM8K tasks*. Given deterministic decoding and a deterministic suppression rule, the same input questions must produce the same conversation trajectories and thus identical accuracy. The non-zero baseline variance (± 1.1 pp) arises because verbose acknowledgments—the primary source of cascade variation—are present in baseline runs but absent in pruned runs; removing them eliminates the downstream variation they cause. Llama's online delta ($+2.0$ pp) is smaller than the offline estimate ($+8.1$ pp), consistent with the warm-start hypothesis discussed in Section 6.

Acknowledgment cascade transition probabilities. We compute $P(\text{ack}_t \mid \text{ack}_{t-1})$—the probability that message t is an acknowledgment given that the preceding message $t{-}1$ was also an acknowledgment—across consecutive message pairs in GSM8K round-robin conversations. Qwen exhibits strong cascading: $P(\text{ack} \mid \text{ack}) = 90.7\%$ ($n = 408$ consecutive ack pairs out of 450 ack-preceded messages). Llama's cascade rate is lower: $P(\text{ack} \mid \text{ack}) = 61.9\%$ ($n = 369$ out of 596), consistent with its undifferentiated communication profile (§5.3). Gemma produces too few acknowledgments ($< 5\%$ of messages; $P(\text{ack} \mid \text{ack}) = 9.7\%$, $n = 72$) for meaningful transition estimation. The backbone-dependent cascade rates (61.9–90.7%) are consistent with aggregate pruning dissolving dependency chains: even at Llama's lower rate, the majority of ack sequences are self-reinforcing, making their collective removal structurally coherent rather than disruptive. Under stochastic decoding (t=0.7), Qwen cascade probability remains stable: $P(\text{ack} \mid \text{ack}) = 92.2\%$ ($n = 94/102$, $N = 99$ GPT-4o-annotated conversations) vs. 90.7% at t=0, indicating that cascade dynamics are robust to temperature variation.

Cascade rate significance test. The observed cascade rates are significantly higher than the null model $P(\text{ack})^2$ (independence assumption). Under H_0: $P(\text{ack}_t \mid \text{ack}_{t-1}) = P(\text{ack})$, we test whether the conditional probability exceeds the marginal rate. For Qwen, $P(\text{ack}) = 47.9\%$ (719/1,500), yielding $P(\text{ack})^2 = 23.0\%$ under independence; the observed $P(\text{ack} \mid \text{ack}) = 90.7\%$ rejects H_0 (chi-squared $\chi^2 = 329.3$, df $= 1$, $p < 10^{-72}$; exact binomial $p < 10^{-72}$). The odds ratio of consecutive ack vs. ack-following-non-ack is 13.7, with $P(\text{ack} \mid \text{non-ack}) = 41.5\%$ ($n = 750$). For Llama, $P(\text{ack}) = 47.0\%$ $(691/1{,}470)$, yielding $P(\text{ack})^2 = 22.1\%$; observed $P(\text{ack} \mid \text{ack}) = 61.9\%$ also rejects H_0 ($\chi^2 = 53.2$, $p = 3.1 \times 10^{-13}$; exact binomial $p = 1.9 \times 10^{-13}$), though the effect is weaker (odds ratio $= 1.30$, $P(\text{ack} \mid \text{non-ack}) = 55.5\%$, $n = 580$). These results confirm that acknowledgment cascading is a genuine sequential dependency, not an artifact of high marginal ack rates.

Position-controlled cascade analysis. The observed cascade rates ($P(\text{ack}_t \mid \text{ack}_{t-1})$) could in principle reflect a position artifact: if later positions are inherently more likely to produce acknowledgments, sequential dependence would be confounded with position. We fit a logistic regression $P(\text{ack}_t) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot \text{ack}_{t-1} + \beta_2 \cdot \text{position}_t)$ to test whether the previous-ack effect survives position control.

For Qwen ($n = 1{,}200$ observations, 300 tasks), prev_ack remains highly significant after controlling for position: OR $= 6.43$ (95% CI $[4.38, 9.43]$, $p = 2.1 \times 10^{-21}$). Position is also significant (OR $= 1.95$, $p = 1.2 \times 10^{-19}$), but the full model (AIC $= 1{,}212.6$) substantially outperforms the position-only model (AIC $= 1{,}315.5$; ΔAIC $= -102.8$).

For Llama ($n = 1{,}176$ observations, 294 tasks), prev_ack is significant with a smaller effect: OR $= 2.02$ (95% CI $[1.49, 2.75]$, $p = 6.5 \times 10^{-6}$). Position shows OR $= 0.73$ ($p = 8.3 \times 10^{-6}$), indicating a *decreasing* ack probability at later positions—opposite to Qwen. The full model (AIC $= 1{,}574.8$) again outperforms position-only (AIC $= 1{,}593.6$; ΔAIC $= -18.9$).

Both backbones confirm that acknowledgment cascading is a genuine sequential dependency, not a position artifact. The stronger Qwen effect (OR $= 6.43$ vs. 2.02) is consistent with Qwen's higher cascade rate (90.7% vs. 61.9%) and function-organized communication profile.

Position effect divergence and suppressor dynamics. The two backbones exhibit opposite position effects: Qwen's position OR $= 1.95$ ($p = 1.2 \times 10^{-19}$) indicates that later rounds produce *more* acknowledgments, consistent with self-reinforcing cascades that intensify over the conversation. Llama's position OR $= 0.73$ ($p = 8.3 \times 10^{-6}$) shows the reverse—acknowledgment probability *decreases* at later positions, indicating cascades that attenuate rather than amplify. This divergence produces a suppressor variable effect in Llama: the bivariate OR without position control is 1.30, but conditioning on position in the full model *increases* the prev_ack OR to 2.02. Position acts as a suppressor because Llama's declining overall ack rate at later rounds masks the true within-position sequential dependency; once this confound is partialled out, the cascade effect strengthens. In contrast, Qwen's prev_ack effect decreases only modestly when position is added (from OR $= 13.7$ unconditional to OR $= 6.43$ controlled), because position and prev_ack reinforce rather than oppose each other. Despite the opposite temporal dynamics—self-amplifying for Qwen, self-dampening for Llama—prev_ack is highly significant in both models ($p < 10^{-5}$), confirming that acknowledgment cascading reflects genuine sequential dependency rather than a position artifact across architecturally distinct backbones.

Task-level cascade rate vs. pruning outcome. We test whether per-task cascade rate predicts whether oracle pruning changes the task outcome (correct→wrong or wrong→correct). For each task in the oracle experiments, we compute the cascade rate (fraction of ack-preceded messages that are also ack) and record whether pruning changed correctness. Qwen ($N = 500$, oracle 5-class): 10 tasks changed outcome (7 improved, 3 degraded); point-biserial $r = -0.047$, $p = 0.297$; Mann-Whitney $U = 2{,}018$, $p = 0.289$. Llama ($N = 500$): 24 tasks changed (12 improved, 12 degraded); point-biserial $r = -0.018$, $p = 0.692$; Mann-Whitney $U = 5{,}100$, $p = 0.707$. Logistic regression yields non-significant cascade coefficients for both backbones (Qwen: $\beta = -1.08$, $p = 0.299$; Llama: $\beta = -0.18$, $p = 0.691$). The null relationship neither supports nor refutes the cascade dissolution hypothesis: the per-task cascade rate does not predict individual pruning vulnerability, which is agnostic to whether aggregate removal succeeds because of collective redundancy or for other reasons.

Cascade length and pruning safety. We stratify oracle pruning outcomes by cascade length—the number of consecutive acknowledgments preceding and including the pruned message—to test whether longer cascades predict safer pruning. Table 31 reports per-bucket safety rates (fraction of tasks where accuracy is unchanged after pruning) across 10 seeds.

Table 31: Pruning safety rate (%) by cascade length (consecutive ack sequence length). N: number of task$\times$seed observations per bucket. Cochran–Armitage trend test assesses monotonic association between cascade length and safety.

Cascade length	**0**	**1**	**2**	**3**	**4+**
Qwen safety (%)	100.0	98.4	98.9	100.0	98.2
n	9	507	1,188	23	272
Llama safety (%)	86.7	92.1	96.5	84.0	96.4
n	90	1,167	463	50	169

Qwen: Cochran–Armitage $z = 1.77$, $p = 0.077$ (NS). Llama: $z = -3.66$, $p = 0.0002$. Llama bucket 3 ($n = 50$, 84.0%) reflects 8 *improved* tasks and 0 degraded; the non-monotonicity is driven by improvements, not failures.

Qwen exhibits uniformly high safety (98.2–100.0%) with no trend ($p = 0.077$), consistent with its function-organized communication where acknowledgments are informationally redundant regardless of sequential context. Llama shows a significant dose-response: safety increases from 86.7% (isolated messages, cascade length 0) to 96.4% (cascades of 4+ consecutive acks; Cochran–Armitage $p = 0.0002$). This provides quantitative support for the cascade dissolution hypothesis (§6): longer acknowledgment sequences exhibit greater internal redundancy, making their collective removal progressively safer.

S.14 Per-Position Label Concordance

We measure whether oracle labels at a given conversation position are stable across independent runs. Using the 10-seed Llama oracle replication ($N = 195$ tasks with valid traces across all seeds, 5 rounds each, 10 seeds), we compute per-position concordance: for each (task, round) pair, we record the function type assigned by each seed's oracle annotator and measure agreement.

Table 32: Per-position label concordance across 10 oracle seeds (Llama/GSM8K, $N = 195$ tasks with valid traces in all seeds). Majority concordance: fraction of seeds agreeing with the modal label. Fleiss' κ: inter-seed agreement on binary ack/non-ack classification. Computed from the 10-seed oracle output files.

	5-class		**Binary (ack/non-ack)**		
Round	Mean conc.	Perfect (%)	Mean conc.	Fleiss' κ	p_{ack}
1	0.761	5.5	1.000	1.000	0.000
2	0.528	0.5	0.628	0.002	0.497
3	0.748	6.5	0.776	0.007	0.228
4	0.613	0.0	0.648	−0.006	0.603
5	0.536	0.5	0.609	−0.017	0.499
Overall	0.637	2.6	0.732	—	—

Round 1 shows perfect binary concordance ($\kappa = 1.0$) because initial solver messages are never acknowledgments ($p_{\text{ack}} = 0$). Rounds 2–5 show near-zero Fleiss' κ (-0.017 to 0.007), indicating that cross-seed agreement on ack classification is at chance level after adjusting for marginal rates. The moderate majority concordance (0.53–0.75 for 5-class) is driven primarily by marginal label frequencies, not genuine position-label coupling. This trajectory dependence—the same position receives different labels under different conversation realizations—supports the cascade dissolution hypothesis: oracle labels are determined by the conversation trajectory, not by position alone, and aggregate removal is safe, consistent with ack sequences being self-reinforcing chains rather than position-fixed roles.

S.15 Cross-Model Ack Concordance

We extend the within-model concordance analysis to test whether acknowledgment placement is task-driven (same questions elicit acks at the same positions regardless of backbone) or model-specific. Using Qwen and Llama oracle annotations on the same 200 GSM8K questions (seed 42), we compute Cohen's κ for the binary ack/non-ack label at each matched (task, round) pair.

Table 33: Cross-model (Qwen vs. Llama) ack concordance on GSM8K ($N = 200$ tasks, seed 42). Pooled κ includes all rounds; stratified κ averages within-round κ weighted by n, removing positional confounds. Round 1 excluded (always non-ack for both models).

Round	n	Qwen p_{ack}	Llama p_{ack}	κ	Observed agr.
2	193	0.990	0.451	−0.021	0.440
3	193	0.140	0.259	0.032	0.684
4	193	0.933	0.617	0.052	0.622
5	193	0.746	0.560	0.009	0.534
Stratified	772	—	—	0.018	—
Pooled	772	0.702	0.472	0.159	—

The stratified $\kappa = 0.018$ (bootstrap 95% CI: $[-0.038, 0.076]$) indicates chance-level agreement once position is controlled. The higher pooled $\kappa = 0.159$ is entirely attributable to shared positional structure (both models show higher ack rates on even rounds), not to task-level concordance. This confirms that acknowledgment placement is model-specific: different backbones produce acks at different questions even when solving the same problems, consistent with the backbone-dominance finding ($V = 0.45$ vs. 0.18; §5.4). Task-level ack proportion correlation is similarly negligible (Pearson $r = 0.09$, $p > 0.2$).

Table 34: Comparison of MAS communication pruning methods. **Granularity**: what is pruned. **Online**: works at inference time without training. **Taxonomy**: uses message-function categories. **Domains**: number of tested task domains. **Stats**: formal statistical testing (McNemar, ANOVA, CI).

Method	**Granularity**	**Online**	**Taxonomy**	**Domains**	**Stats**
AgentDropout	Agent+edge	×	×	6	×
AgentPrune	Edge	×	×	6	×
AGP	Agent+edge	×	×	3	×
S^2-MAD	Turn	✓	×	3	×
Ours	Message	✓	✓	2+2†	✓

†2 validated (GSM8K, StratQA; multi-seed, dual backbone) + 1 supplementary (ARC: single-seed offline, N=250) + 1 boundary condition (MATH: -2.0 pp at N=200; pilot -10 pp at N=50).

S.16 Practical Cost Savings

In online deployment, suppressed messages are excluded from the context window of subsequent turns, compounding input-token savings over multi-round conversations. Under greedy decoding on GSM8K (6 rounds, 2 agents), Qwen suppresses 22% of agent messages and Llama suppresses 75%, reducing cumulative input tokens by approximately 15% and 55%, respectively. The primary practical value is latency: fewer context tokens translate directly to lower time-to-first-token in autoregressive serving, with Llama's 75% message suppression translating to proportional latency reduction. Token savings scale proportionally with serving cost under any pricing model.

Complementarity with other efficiency methods. Ack pruning operates at the within-turn message level, complementing agent-level methods (AgentDropout, AgentPrune: 20–90% reduction by removing entire agents) and turn-level methods (S^2-MAD: early stopping). Our method targets redundancy that persists even when all agents and turns are preserved, and can in principle be stacked with agent-level or turn-level pruning for compounding savings—though this composition remains untested.

S.17 Llama Pruning Mechanism (Hypothesis)

We hypothesize two backbone-divergent pruning-safety mechanisms. Qwen acknowledgments are informationally redundant (bigram novelty 5.7%, median token overlap 1.0; Appendix T). Llama acknowledgments carry more content (mean 59.8 tokens vs. Qwen's 10.7) and exhibit a one-directional error channel: 0% error correction but 14.3% error injection. Removing them may reduce error propagation while sacrificing no corrective information. These patterns are consistent with the dual-mechanism hypothesis but do not constitute confirmation; controlled interventions (e.g., injecting synthetic acknowledgments with and without errors) are needed to establish causality.

S.18 Per-Type κ Impact Analysis

The low information_provision κ (≈ 0.237) affects different analyses to different degrees. We distinguish three tiers:

1. **Unaffected**: Ack pruning safety, perturbation–sensitivity dissociation, and aggregate F/LRT tests. These depend on the acknowledgment category ($\kappa > 0.8$) and omnibus between-type variance, not on individual non-ack labels.
2. **Usable with caveat**: OLS per-type β coefficients for information_provision. Systematic information_provision$\leftrightarrow$verification confusion may redistribute explained variance between these two categories without altering the overall model fit or the acknowledgment coefficient.
3. **Directly compromised**: FG pruning comparisons and per-type rankings that depend on information_provision labels. Because the heuristic detector does not use function-type labels, the practical pruning pipeline is unaffected.

S.19 Cohen's h Effect Sizes

We compute Cohen's $h = 2 \arcsin\sqrt{p_{\text{pruned}}} - 2 \arcsin\sqrt{p_{\text{baseline}}}$ [Cohen, 1988] for every experiment with paired accuracy data. Across the 23 non-boundary conditions in Table 41, $|h|$ ranges from 0.000 (72B heuristic, Qwen/ARC) to 0.111 (Llama/StratQA oracle)—all below the 0.20 "small" threshold. At N=200 (102 valid), MATH yields $h = -0.048$ (negligible); the Qwen3-8B boundary condition reaches "small" ($h = -0.380$). For the 72B N=100 zero-flip probe and the Qwen/ARC offline result, $h = 0$ exactly. The uniformly negligible effect sizes complement the McNemar p-values by confirming that even the non-significant point estimates represent practically zero-magnitude effects, not merely underpowered tests.

S.20 Token-Count-Matched Random Pruning

The message-count-matched random baseline prunes the same *number* of messages as ack-only but not the same volume of content. Because acknowledgments are short (Qwen: $\sim$12 tokens vs. $\sim$64 overall; Llama: $\sim$80 vs. $\sim$110), count-matched random removes substantially more tokens ($5.5\times$ for Qwen, $1.4\times$ for Llama). We test whether this asymmetry explains the accuracy gap by constraining random pruning to match ack-only's *token* budget (greedy knapsack, 10 seeds).

Table 35: Offline accuracy risk (%) under three pruning regimes. Risk = fraction of correct tasks whose final answer changes (10 seeds, mean $\pm$ SD for random). Gemma results based on $n = 11$ ack messages; interpret with caution.

Backbone	Ack-only	Count-matched	Token-matched	Token ratio
Qwen	2.6	1.2±0.3	0.2±0.2	5.51×
Llama	1.8	1.2±0.6	0.8±0.6	1.38×
Gemma	0.0	0.0±0.0	0.0±0.0	8.14×

Token-matched random achieves lower risk than ack-only (Qwen: 0.2% vs. 2.6%; Llama: 0.8% vs. 1.8%), suggesting the e2e accuracy gap (Table 23: ack 92% vs. random 87.3%) is primarily driven by the $\sim$$5\times$ token asymmetry. However, this offline risk simulation assumes message independence; ack-only's practical value lies in *interpretability*—targeting a semantically coherent class with low information content—rather than raw accuracy advantage over token-matched random.

S.21 Extended Limitations

AWQ quantization. The 72B evaluation uses AWQ 4-bit quantization [Lin et al., 2024]. Comparable acknowledgment suppression rates between 72B-AWQ (66.8%) and full-precision 7B (56.3%) provide indirect evidence that quantization does not fundamentally alter communication patterns.

Stochastic decoding. Under t=0.7, Llama oracle $\Delta = -2.2$ pp (Fisher $p = 0.29$, NS) while Qwen remains consistently safe across 10 seeds ($\Delta = +0.30 \pm 1.80$ pp, TOST $p < 0.001$ at $\delta = -3$ pp; Appendix C). Llama oracle suppression drops from 36.6% to 28.0%, consistent with fewer cascade-redundant acknowledgments when sampling introduces response diversity.

Inter-rater reliability. Human inter-rater $\kappa = 0.76$ ($N = 225$, same 5-class protocol; Appendix S.18); core claims depend on ack ($\kappa > 0.8$). The zero error-correction rate (0/1,483 messages) combined with flat dose-response curves provides indirect evidence against hidden quality degradation beyond task accuracy.

Online oracle annotator agreement. The GPT-4o-mini online annotator was validated against GPT-4o on a 50-sample subset (Cohen's $\kappa = 0.822$, 95% CI $[0.70, 0.94]$). Of 7 disagreements, 3 involved reasoning_guidance $\leftrightarrow$ information_provision confusions—the same boundary that produces the lowest per-type κ in cross-model validation (§S.18). The remaining 4 disagreements were distributed across other category pairs. Crucially, no disagreements involved the acknowledgment category, consistent with the high per-type ack $\kappa > 0.8$ and preserving the validity of ack-based pruning decisions in the online pipeline.

Discarded N=1,000 Llama oracle run. A preliminary N=1,000 Llama oracle run showed anomalous suppress rate (74.5% vs. expected $\sim$37%), indicating GPT-4o over-classification at larger scale. This run was discarded; all reported Llama oracle results use the validated N=500 and N=200 configurations.

FG offline replay. FG comparisons rely on offline replay; online token-matched random confirms the finding (efficiency: Appendix W). The impact gradient breaks down under selector topology. Three-agent evidence is preliminary (Appendix X).

Heuristic detector calibration. The heuristic detector's character-length threshold achieves F1 = 0.845 on GSM8K vs. 0.213 on StratQA for Qwen, indicating substantial domain-specific calibration requirements.

Oracle distribution shift. Oracle labels computed on baseline conversations face a distribution-shift caveat: online pruning alters trajectories from the first suppression point, potentially rendering later-position labels stale. Empirically, this gap is negligible for Qwen (offline $+1.0$ pp vs. online $+0.8$ pp) but larger for Llama (offline $+8.1$ pp ($N = 297$, single-seed) vs. online $+2.1 \pm 2.4$ pp (10-seed mean)), suggesting attenuation from distribution shift, sample-size differences, or both (Appendix S.12).

Replication asymmetry. Both backbones have 10-seed oracle replication (Qwen: $\Delta = +0.40 \pm 1.02$ pp; Llama: $\Delta = +2.1 \pm 2.4$ pp); however, Llama's larger seed-level variance (SD = 2.4 vs. 1.0 pp) yields wider CIs, and the heuristic replication is asymmetric (Qwen 10 seeds, Llama 5 seeds). Heuristic strategy comparisons (oracle vs. heuristic vs. length-only) are concentrated on GSM8K; StrategyQA and ARC validate the dissociation phenomenon and oracle pruning generalization but do not independently validate heuristic detector performance across domains.

Oracle API dependency. The oracle pipeline depends on a proprietary LLM API (GPT-4o/GPT-4o-mini): annotation cost scales linearly with message volume ($13.47 for 5,388 messages with GPT-4o), and availability is subject to rate limits and service continuity. Cross-model annotation agreement ($\kappa = 0.822$) is not perfect; residual label noise propagates into both offline and online pruning decisions. The offline pipeline assumes baseline conversation trajectories cover the relevant message space—an assumption that weakens in open-ended dialogue settings where pruning-induced trajectory divergence is more severe.

Heuristic non-transferability. The model-adapted heuristic requires per-backbone rule design—a practical limitation analogous to task-specific prompt engineering. While the rules are simple (character thresholds, keyword lists, repetition ratios), they do not transfer across backbone families without recalibration (§5.5). This positions the heuristic as a proof-of-concept demonstrating that annotation-free pruning is feasible, not as a universal detector.

S.22 Extended Discussion

Length as proxy feature: convergence and divergence. Direct comparison of length-only and composite heuristic pruning quantifies the extent to which message length alone can serve as a proxy for acknowledgment detection. On Qwen2.5-7B/GSM8K, the two detectors converge: length-only pruning yields $\Delta = +0.50 \pm 1.50$ pp with a 42.2% suppress rate (3 seeds), while the composite heuristic yields $\Delta = -0.05 \pm 0.99$ pp with a $\approx 42\%$ suppress rate (10 seeds; Table 2). Both accuracy deltas are statistically indistinguishable from zero, and the suppress-rate overlap indicates that the additional heuristic features—keyword matching, repetition detection—capture few messages beyond those already identified by the length threshold alone. Length thus constitutes not merely a correlated feature but the *primary discriminative axis* for Qwen acknowledgments, consistent with their morphologically concentrated surface form (median 4 tokens; Appendix T). The detectors diverge sharply on Llama-3.1-8B/GSM8K: length-only pruning suppresses only 0.76% of messages—a near-vacuous operation ($\Delta = -0.10 \pm 1.29$ pp, 5 seeds)—whereas the composite heuristic suppresses $\sim 75\%$ of messages (vs. $\sim 37\%$ oracle ack rate) with $\Delta = +2.20 \pm 1.44$ pp (non-inferiority $p < 0.001$; 5 seeds; Table 2)—a $\sim 99\times$ gap in suppression rate that underscores the need for pattern-level features beyond length. The near-zero length-only suppress rate confirms that Llama's acknowledgments rarely fall below the character threshold; the near-zero accuracy delta (-0.10 pp) reflects the vacuity of the filter rather than any safety property, since 99.2% of messages pass through unchanged. This pattern reflects a fundamental difference in how backbones distribute communicative content across message lengths: Qwen concentrates acknowledgments in a narrow, low-token surface band amenable to a single threshold, whereas Llama embeds functionally equivalent content within verbose, lexically diverse outputs (65.6% bigram novelty; Appendix T), requiring pattern-level features for safe identification. The asymmetry implies that effective pruning detectors must be calibrated to backbone-specific surface morphology; no single-threshold mechanism can serve as a universal pruning rule across architectures.

Suggestive positive trend under heuristic pruning. Across both oracle and heuristic pruning on Llama/GSM8K, point estimates are consistently non-negative (oracle 10-seed $\Delta = +2.1 \pm 2.4$ pp, 9/10 positive; heuristic 5-seed $\Delta = +2.20 \pm 1.44$ pp, 5/5 positive, 95% CI $[+0.41, +3.99]$ pp), suggesting that removing redundant acknowledgments may reduce downstream reasoning drift—the tendency for agents to echo or elaborate on checker confirmations rather than advancing the solution. The heuristic's over-suppression ($\sim 75\%$ vs. $\sim 37\%$ oracle ack rate) implies that $\sim 38\%$ of suppressed messages are non-acknowledgments; the positive accuracy delta despite this over-suppression is consistent with Llama's one-directional error channel (0% correction, 14.3% injection; Appendix S.17)—removing error-injecting messages and their cascading effects outweighs any cost of suppressing a minority of informative messages. However, the current sample size is insufficient for causal inference on effect direction: non-inferiority is established at $\delta = -3$ pp ($t = 8.07$, $p < 0.001$); a separate one-sided test for positive effect yields $t = 3.42$, $p = 0.013$, $df = 4$, but a positive treatment effect would require a separate superiority test with substantially more power. We flag this as an exploratory finding warranting future investigation with larger N and controlled manipulation of ack content.

S.23 Heuristic Threshold Sensitivity

Threshold sensitivity analysis. We vary the heuristic detector's threshold parameters to assess whether pruning safety depends on specific threshold choices. Table 36 reports pruned accuracy across four detector variants on Llama-3.1-8B-Instruct / GSM8K ($N = 200$, seed $= 42$, greedy decoding, vLLM inference).

Pruned accuracy is stable across tested configurations (81.5–82.5%, ≤ 1.0 pp range) despite substantial variation in suppression intensity (23.1–75.2%). The keyword-only variant suppresses only 23.1% of messages yet achieves the same pruned accuracy as the more aggressive length-based variants,

Table 36: Heuristic threshold sensitivity (Llama/GSM8K, $N = 200$, single seed). Pruned accuracy is stable across detector configurations (81.5–82.5%). Baseline variability (77.7–81.5%) reflects vLLM inference non-determinism for V1–V3 (± 2 pp); V4's lower baseline (77.7%) additionally reflects sample bias from 35% data loss.[†]

Variant	**Configuration**	**Pruned Acc (%)**	**Suppress (%)**	**Baseline (%)**
char<200	LEN=200, default kw+ngram	82.5	75.2	81.5
char<400	LEN=400, default kw+ngram	82.0	76.2	80.0
keyword-only	LEN=0, OVERLAP=999	82.5	23.1	79.5
ngram+length	KW=[], ngram+length only	81.5[‡]	73.4	77.7[‡]

[†]Each variant uses a separate baseline run; V1–V3 baseline differences (± 2 pp) are consistent with vLLM inference non-determinism observed across independent runs under identical configurations (cf. multi-run variance analysis above).
[‡]130/200 tasks completed; 70 tasks lost to vLLM infrastructure failure, introducing sample bias beyond inference non-determinism.

confirming that the heuristic's safety margin is not an artifact of a specific length threshold. This is a single-seed exploratory analysis; see Section 5.2 for multi-seed evidence under the default heuristic configuration.

T Information-Theoretic Analysis of Acknowledgments

To quantify why acknowledgment pruning under greedy decoding does not significantly degrade accuracy, we measure five information-theoretic properties of each message relative to its predecessor, grouped by communication function type. Metrics are computed on Qwen and Llama round-robin traces (300 tasks each); Gemma is excluded due to insufficient acknowledgment count ($n = 11$).

Table 37: Information-theoretic properties by function type (Qwen/GSM8K/RR, $N = 1{,}500$ messages). All ack vs. non-ack comparisons significant at $p < 10^{-10}$ (Mann–Whitney U) except where noted.

Metric	**Ack**	**Info Prov.**	**Reasoning**	**Verif.**	**Coord.**
Bigram novelty	**0.057**	0.317	0.749	0.434	—
Token length (median)	**4**	95	117	53	—
Jaccard overlap	**0.686**	0.653	0.313	0.482	—
Semantic similarity	**0.783**	0.896	0.823	0.841	—
Information density	**0.971**	0.565	0.566	0.685	—

Table 38: Information-theoretic properties by function type (Llama/GSM8K/RR, $N = 1{,}470$ messages).

Metric	**Ack**	**Info Prov.**	**Reasoning**	**Coord.**	**Verif.**
Bigram novelty	0.656	0.759	0.723	0.925	0.625
Token length (mean)	59.8	120.1	124.8	72.4	51.5
Jaccard overlap	0.342	0.332	0.336	0.191	0.283
Semantic similarity	0.636	0.817	0.763	0.453	0.706
Information density	**0.774**	0.472	0.519	0.741	0.718

Surface-level information content. Table 39 complements the above metrics with lexical diversity (type–token ratio, TTR) and bigram/trigram overlap with the immediately preceding message, computed over word-level tokens. Across both backbones, acknowledgments exhibit the highest n-gram overlap with their predecessors (Qwen: 0.757 bigram, 0.733 trigram; Llama: 0.426, 0.363), confirming that they primarily echo prior content. Conversely, information provision and coordination messages show near-zero overlap (< 0.11 for Llama; < 0.26 for Qwen), indicating genuinely novel content. This quantitative gap directly explains the perturbation-sensitivity dissociation: perturbing an acknowledgment alters surface form but not functional information, since the replaced content was redundant with the preceding turn.

Table 39: Surface-level information metrics by function type and backbone (GSM8K/round-robin). *Bi-OL* and *Tri-OL*: fraction of current-message bigrams/trigrams also present in the preceding message. TTR: type–token ratio (unique words / total words). Gemma excluded ($n_{\text{ack}} = 11$).

Backbone	**Function Type**	N	**Tokens**	**TTR**	**Bi-OL**	**Tri-OL**
Qwen	Acknowledgment	951	39.2	0.820	**0.757**	**0.733**
	Verification	106	86.7	0.529	0.859	0.813
	Information provision	82	99.9	0.520	0.255	0.185
	Reasoning guidance	361	59.2	0.654	0.349	0.297
Llama	Acknowledgment	670	82.7	0.595	**0.426**	**0.363**
	Verification	112	115.8	0.422	0.420	0.369
	Information provision	192	62.1	0.665	0.235	0.184
	Reasoning guidance	376	69.2	0.624	0.168	0.105
	Coordination	120	37.4	0.845	0.105	0.064

Dual mechanism. Qwen acknowledgments are informationally vacuous: median bigram novelty is 0% and median Jaccard overlap with the preceding message is 1.0, meaning they are near-verbatim echoes. Pruning removes no unique information. Llama acknowledgments are longer and carry more novel content (bigram novelty 65.6%), but their pruning benefit arises from a different mechanism: they inject errors at 14.3% with 0% correction, so removal cuts an error-propagation channel rather than removing redundancy.

Per-round acknowledgment trends. The proportion of acknowledgments varies across conversation rounds and backbones. Qwen shows monotonic increase from 0% (round 1) to 96% (round 5), consistent with later rounds serving primarily confirmatory roles. Llama exhibits an inverted-U pattern peaking at 81% in round 3 before declining, suggesting a transition from acknowledgment to substantive re-engagement in later rounds. Gemma produces near-zero acknowledgments across all rounds, consistent with its position-organized profile where the acknowledgment category is functionally absent.

T.1 Heuristic Detector Performance

Backbone-specific heuristic detectors using only deterministic features (length, keywords, n-gram repetition) achieve F1 $= 0.845$ (Qwen, rule: character length < 30) and 0.805 (Llama, composite rule: keyword matching $+$ length $< 300 + 4$-gram repetition > 0.3) with 0–1 pp pruning gap versus oracle labels, validated via 5-fold GroupKFold CV (grouped by task). Composite rules generalize cross-domain: Llama F1 > 0.71 on StratQA and ARC. Qwen's single-threshold rule degrades cross-domain (F1 $= 0.21$ on StratQA), motivating the multi-feature approach for deployment. Full development details appear in §4.6.

U Scale Validation: Qwen2.5-14B

We extend the scale trajectory analysis (Section 5) with detailed results for Qwen2.5-14B-Instruct on GSM8K under round-robin topology, bridging the 7B and 72B endpoints.

U.1 Function Annotation

GPT-4o annotation of 600 agent messages yields an acknowledgment rate of 31.7% (190/600), closely matching the 72B relabeling rate (29.7%). The full function distribution is: VERIFICATION 191, ACKNOWLEDGMENT 190, REASONING_GUIDANCE 162, INFORMATION_PROVISION 57, COORDINATION 0. The near-identical acknowledgment proportions across 14B and 72B suggest that communication function distributions are stable within the Qwen2.5 family at these scales.

Cross-model annotation agreement. Qwen2.5-72B-Instruct-AWQ relabeling yields an acknowledgment rate of 29.7% (178/600), within 2 pp of the GPT-4o rate (31.7%), confirming annotator consistency at this model scale.

U.2 Heuristic Detector

A character-length threshold of 80 achieves F1 $= 0.837$ (precision $= 0.799$, recall $= 0.879$) against GPT-4o labels. The higher threshold compared to 7B (char_threshold $= 30$) reflects 14B's tendency to produce longer acknowledgment messages. Despite the different threshold, classification quality remains comparable to the 7B detector (F1 $= 0.845$).

U.3 Simulate Pruning

Offline simulate pruning based on 72B annotations (which closely agree with GPT-4o) yields: baseline accuracy 95% (from offline traces), ack-only accuracy 95%, $\Delta = 0\,$pp, with 18.3% token savings. The zero delta is consistent with acknowledgment content carrying no marginal task-relevant information at 14B scale, matching the 7B finding.

U.4 Online Pruning

Online pruning (skip mode, char_threshold $= 80$, $N = 100$) yields: baseline 90%, ack-only 93%, $\Delta = +3\,$pp, with 5.8% suppression rate (29/500 messages). The low suppression rate—substantially below the annotated acknowledgment proportion (31.7%)—reflects the conservative char_threshold: at threshold $= 80$, only the shortest acknowledgments are suppressed online, while many genuine acknowledgments exceed 80 characters and pass through.

Unified threshold validation. To enable fair cross-scale comparison, we re-run online pruning with char_threshold $= 30$ (matching the 7B and 72B threshold). This yields: baseline 91%, ack-only 95%, $\Delta = +4\,$pp, with 6.4% suppression rate (32/500 messages). The minimal increase from 5.8% to 6.4% indicates that the low suppression rate is intrinsic to 14B's acknowledgment length distribution—14B generates few short (< 30 character) acknowledgments—rather than an artifact of threshold calibration. The heuristic thus captures only $\sim$20% of true acknowledgments at 14B (6.4% suppressed vs. 31.7% true ack rate per GPT-4o), illustrating that length-based heuristic transferability varies by scale. This makes the 14B online result a test of partial heuristic coverage rather than a full pruning safety test; the simulate pruning result ($\Delta = 0\,$pp with oracle labels) provides the complementary full-removal evidence.

Interpreting the low suppression rate. Both thresholds yield $< 7\%$ suppression, meaning the delta is based on removing only $\sim$30 messages out of 500. While the direction is consistently positive (no degradation), the small number of suppressed messages limits the strength of inference about what would happen under more aggressive pruning. The simulate pruning result ($\Delta = 0\,$pp at full acknowledgment removal) provides complementary evidence that complete removal is safe.

Run-to-run variance. Under greedy decoding, single-run point estimates carry $\pm$1–2 pp residual variance from GPU numerical precision (cf. ten-run replication for 7B: $\pm 0.99\,$pp). The $+3\,$pp and $+4\,$pp deltas are within this variance band and should not be interpreted as genuine improvements.

U.5 Full-Scale 72B Preliminary Probe ($N = 265$)

The exploratory 72B probe ($N = 100$, $\Delta = 0\,$pp) uses a sample too small for powered inference. We extend to $N = 265$ with oracle (GPT-4o 5-class) labels to provide a preliminary single-seed scale test.

Online oracle pruning. Qwen2.5-72B-Instruct-AWQ on GSM8K ($N = 265$): baseline 92.1%, pruned 92.1%, $\Delta = +0.0\,$pp ($p = 1.000$), with 66.8% message suppression. The suppression rate at 72B (66.8%) can be compared to 7B (56.3%) to assess whether larger models produce proportionally different acknowledgment rates.

Cross-scale comparison. Table 40 summarizes oracle pruning deltas across the Qwen2.5 scale trajectory.

Table 40: Pruning Δ (pp) across Qwen2.5 scales on GSM8K (oracle where available, heuristic otherwise; greedy decoding).

Scale	N	**Baseline**	**Pruned**	Δ	**Supp.**
7B	500	88.8	89.6	$+0.8$	56.3%
14B†	100	90/91	93/95	$+3/+4$	5.8/6.4%
72B	100	—	—	0	—
72B	265	92.1	92.1	$+0.0$	66.8%

† 14B uses char-length heuristic (thresholds 80/30); oracle online pruning was not run at this scale. Offline oracle ($\Delta = 0$ pp) provides complementary full-removal evidence.

U.6 MAS Deployment Positioning

Our study analyzes communication *within* multi-agent systems, not whether multi-agent architectures outperform single-agent alternatives—a question addressed by existing MAS benchmarks Du et al. [2025a], Zhu et al. [2025]. The taxonomy is an empirically-derived analytical framework for understanding inter-agent message functions; its value lies in identifying which messages are safe to remove from an existing MAS pipeline, not in justifying MAS over single-agent approaches. Single-agent baselines are orthogonal: they would test whether agents should communicate at all, whereas we test which communications are redundant given that agents *do* communicate. This distinction parallels neural pruning LeCun et al. [1990], where the question is which weights to remove from a trained network, not whether the architecture should have fewer layers.

U.7 Scale Trajectory of Pruning Safety

Ack-pruning direction is directionally consistent across Qwen2.5 7B and 14B scales, with preliminary single-seed evidence at 72B: 7B ($\Delta = +0.8$ pp, $N = 500$), 14B (offline oracle $\Delta = 0$ pp), and 72B ($N = 265$, $\Delta = +0.0$ pp, zero discordant flips, $p = 1.000$, 66.8% suppression; Table 2; Appendix S.3). No significant degradation is observed at any individual scale point; the 72B result is preliminary evidence (single seed, insufficient to estimate seed-to-seed variance) with $N = 265$ (truncated from 500 due to API rate limits; binomial 95% CI upper bound on flip rate $\approx 1.1\%$). The 14B result is limited by low heuristic suppression (6.4%); the offline oracle ($\Delta = 0$ pp) provides complementary evidence. Individual-message perturbation at 72B remains unmeasured; the dissociation is inferred from 7B perturbation data.

V Cross-Experiment Summary

Table 41 consolidates all oracle and pruning experiments across backbones, scales, domains, and temperatures. Cohen's h quantifies effect size on the arcsine-difference scale; all non-boundary $|h| < 0.20$, falling in the "negligible" range by conventional thresholds [Cohen, 1988]. For the 72B zero-flip result, $h = 0$ exactly—the strongest possible "no effect" evidence.

W Efficiency Analysis

Computational efficiency. The primary efficiency gain from acknowledgment pruning is *inference call reduction*: each suppressed message eliminates one full LLM generation call. Oracle-guided pruning suppresses **56.0%** of inter-agent messages on Qwen2.5-7B (aggregated over ten seeds, N=2,000; **2.80 per task**) and **36.5%** on Llama-3.1-8B (ten seeds, N=1,939; **1.84 per task**), translating directly to proportional wall-time savings under roughly uniform per-round latency (Table 42). Token savings are substantially lower—approximately **10.1%** on Qwen and **27.5%** on Llama—because acknowledgments are much shorter than substantive messages (Qwen ack median: 4 tokens vs. verification/reasoning: 53–117 tokens). The disparity is itself informative: *ack pruning's value lies in eliminating redundant inference calls, not in reducing token volume*. In deployment, backbone-specific heuristic detectors (F1 = 0.845 for Qwen, 0.805 for Llama; §4.6) achieve these savings at zero marginal cost, requiring no LLM annotation.

Table 41: Full cross-experiment pruning matrix. Δ: pruned $-$ baseline accuracy (pp). p: McNemar χ^2 with continuity correction (single-run) or paired t (multi-seed); exact binomial p-values agree within 0.001 for all entries. TOST: equivalence p at stated margin. h: Cohen's arcsine effect size. Non-boundary $|h| < 0.20$ (negligible); Qwen3-8B reaches "small" ($|h| = 0.38$); MATH at N=200 is negligible ($|h| = 0.048$). †Fisher combined across two independent N=500 runs. ‡Heuristic detector, not oracle. §Simulate pruning (offline with oracle labels). $^{\|}$ Single-seed; TOST requires multi-seed variance estimate.

Backbone	**Domain**	**Temp**	N	**Seeds**	**Protocol**	**Δ (pp)**	p	**TOST**	h
Oracle pruning (GPT-4o 5-class labels)									
Qwen-7B	GSM8K	0	500	1	Oracle	$+0.8$	.343	—	$+.026$
Qwen-7B	GSM8K	0.7	500	1	Oracle	$+0.6$	.710	—	$+.018$
Qwen-7B	StratQA	0	500	1	Oracle	$+0.2$	1.000	—	$+.005$
Llama-8B	GSM8K	0	500	1	Oracle	-0.6	.710	—	$-.016$
Llama-8B	GSM8K	0.7	1,000	2†	Oracle	-2.2	.291	—	$-.054$
Llama-8B	StratQA	0	238	1	Oracle	$+5.0$	.180	—	$+.111$
Qwen-72B	GSM8K	0	265	1	Oracle	$+0.0$	1.000	—$^{\|}$	.000
Multi-seed replication									
Qwen-7B	GSM8K	0	200×10	10	Oracle	$+0.40$	—	±1 pp: .048*	$+.013$
Qwen-7B	GSM8K	0	200×10	10	Heuristic‡	-0.05	.876	±1 pp: .007**	$-.002$
Llama-8B	GSM8K	0	∼194×10	10	Oracle	$+2.08$	.023*	±3 pp: <.001***	$+.057$
Llama-8B	StratQA	0	200×8	8	Heuristic‡	$+2.81$	.084	±3 pp: .002**	$+.060$
Temperature robustness (heuristic detector)									
Qwen-7B	GSM8K	0.7	500	1	Heuristic‡	-0.6	.728	—	$-.019$
Llama-8B	GSM8K	0.7	500	1	Heuristic‡	-3.4	.100	—	$-.084$
Gemma-9B	GSM8K	0.7	100	1	Offline	$+3.0$	.453	—	$+.076$
Scale validation (Qwen2.5 family)									
Qwen-14B	GSM8K	0	100	1	Simulate§	0.0	—	—	.000
Qwen-72B	GSM8K	0	100	1	Heuristic‡	0.0	—	—	.000
Cross-domain offline (ack-only vs. baseline)									
Qwen-7B	ARC	0	250	1	Offline	$+0.8$	—	—	$+.019$
Llama-8B	ARC	0	250	1	Simulate§	$+4.0$	.013*	—	$+.088$
Gemma-9B	GSM8K	0	96	1	Offline	-1.0	—	—	$-.023$
Online heuristic pruning (from-scratch)									
Llama-8B	GSM8K	0	1,000	1	Online	$+2.1$	.004**	—	$+.055$
Qwen-7B	StratQA	0	200×5	5	Heuristic‡	$+0.70$	.160	±3 pp: <.001**	$+.015$
Random baseline									
Qwen-7B	GSM8K	0	500	1	Token-matched	$+0.8$	.343	—	$+.026$
Topology / agent count									
Qwen-7B (3-agent)	GSM8K	0	100	1	Online	-1.0	1.000	—	$-.036$
Boundary conditions (expected degradation)									
Qwen3-8B	GSM8K	0	100	1	Heuristic‡ (char<80)	-14.0	<.03*	—	$-.380$
Qwen-7B	MATH	0	200 (102 v)	1	Online	-2.0	.625	—	$-.048$

Oracle cost and upper-bound positioning. The GPT-4o oracle classifier used for pruning validation incurs **\$13.90–\$14.42 per 1,000 tasks** (3,475–3,605 classification tokens per task at current API pricing), with per-message classification adding latency overhead. This positions oracle pruning as a *research upper bound*: it isolates true acknowledgments with high precision, enabling controlled measurement of pruning safety (§5.2), but its cost and latency make it impractical for deployment. The heuristic detectors (§4.6) replace the oracle at zero cost—simple deterministic rules (Qwen: character length < 30; Llama: keyword + length + repetition) executable in microseconds, with no API dependency. For open-weight models on local infrastructure—the setting of this paper—savings manifest as reduced GPU-seconds and lower round-trip latency; for API-served models, each suppressed message eliminates a full API call whose cost includes per-call overhead beyond raw token pricing.

Table 42: Efficiency of oracle-guided ack pruning on GSM8K. Calls saved = suppression rate; token savings are lower because acks are short (median 4 tokens). Oracle cost is for GPT-4o 5-class classification; heuristic detectors achieve comparable savings at zero cost (§4.6).

Backbone	N	Calls Saved	Msgs / Task	Est. Token Saved	Oracle / Task	Oracle Cost / 1K
Qwen2.5-7B	2,000	56.0%	2.80 / 5.0	∼10.1%	3,475 tok	$13.90
Llama-3.1-8B	1,939	36.5%	1.84 / 5.0	∼27.5%	3,605 tok	$14.42

X Exploratory Multi-Agent and Alternative Topology Pilots

To probe whether acknowledgment patterns generalize beyond two-agent solver-checker configurations, we conduct two exploratory pilots using Qwen2.5-7B-Instruct on GSM8K and StrategyQA ($N = 100$ each, greedy decoding).

Three-agent solver-checker-critic. We extend the standard two-agent pipeline with a third agent (LogicCritic) that reviews the checker's verification for completeness, cycling in round-robin order over 7 rounds. The heuristic detector (char < 30) suppresses 37% of agent messages (222/600), with suppression concentrated in checker and critic roles (Table 43). Pruned accuracy (91%) is within 1 pp of baseline (92%; McNemar $p = 1.0$, 1 discordant pair). The ack distribution across agent positions—19% (solver), 40% (checker), 41% (critic)—suggests that downstream agents produce more acknowledgments, consistent with the cascade dynamics observed in two-agent systems.

Debate (advocate–challenger). We replace the solver-checker topology with symmetric argumentation: an Advocate argues for an answer while a Challenger identifies weaknesses, cycling over 6 rounds on StrategyQA (yes/no). The heuristic detector suppresses only 0.21% of messages (1/485), with 99 of 100 tasks producing zero suppressed messages. Baseline accuracy is 62% (pruned 63%, $\Delta = +1$ pp). The near-zero suppression rate indicates that debate agents produce long argumentative responses rather than short acknowledgment-like messages, suggesting that ack prevalence is topology-dependent rather than a universal property of multi-agent LLM communication. An alternative explanation is that the heuristic (char < 30) is poorly calibrated for debate responses; oracle annotation was not performed for this pilot.

Table 43: Exploratory topology pilots (Qwen2.5-7B, $N = 100$, greedy decoding). Suppression uses the heuristic detector (char < 30); no oracle annotation was performed for the debate pilot.

Topology	Domain	Agents	Baseline	Pruned	Δ (pp)	Suppress %
3-Agent (S-C-Cr)	GSM8K	3	92%	91%	-1.0	37.0%
Debate (A-Ch)	StrategyQA	2	62%	63%	$+1.0$	0.21%

Both pilots are exploratory ($N = 100$, single backbone, heuristic detection only). Definitive claims about multi-agent or alternative-topology pruning require larger-scale validation with oracle annotation.

Y Artifact Description

All code, data, and annotation materials are released under the MIT License at `[anonymized]`.

Y.1 Released Artifacts

Code repository. The repository contains the full experimental pipeline:

- **Trace collection** (`src/trace_collection.py`): multi-agent conversation recording via AG2, supporting round-robin, sequential, and selector topologies.
- **Functional annotation** (`src/functional_annotation.py`, `src/gpt4o_annotation.py`): LLM-based 5-class message function annotation with cross-model agreement.

- **Acknowledgment detection** (`src/heuristic_ack_detector.py`, `scripts/train_ack_classifier.py`): rule-based and trained neural classifiers for identifying acknowledgment messages.
- **Perturbation analysis** (`src/local_impact.py`): per-message remove and neutralize ablations with LDIM computation.
- **Pruning validation** (`src/pruning_validation.py`): offline simulation and online pruning with oracle or heuristic detection.
- **Statistical analysis** (`src/analysis.py`): McNemar tests, TOST equivalence tests, likelihood-ratio tests, and figure generation.

YAML configuration files in `configs/` specify all model, agent, data, and topology parameters for reproducibility.

Datasets.

- **Sample traces** (`data/sample_traces/`): representative annotated traces from one backbone-domain configuration, with complete traces for all configurations available in the repository.
- **Human annotation data** (`data/human_annotation/`): annotation sheets from three independent annotators in CSV format, with inter-annotator agreement scripts (Cohen's κ, Fleiss' κ, Krippendorff's α).
- **Classifier training data** (`data/classifier/`): train/validation/test splits for acknowledgment classifier training.

Annotation materials. The repository includes the annotation guideline document (`docs/annotation_guide.md`) and a standalone web-based annotation interface (`data/human_annotation/annotator_standalone.html`) used in the human validation study described in Section 4.

Y.2 Protocol Reuse

Researchers can apply our analysis framework to their own multi-agent systems in three steps:

1. **Collect multi-agent traces.** Record inter-agent message logs from the target system. The pipeline accepts JSONL traces; `src/mast_utils.py` provides utilities for converting from the MAST trace format. For AG2-based systems, `src/trace_collection.py` handles trace recording directly.
2. **Annotate communication functions.** Run `src/functional_annotation.py` to classify each message into the five-type taxonomy (Acknowledgment, Information Provision, Reasoning Guidance, Verification, Coordination). Alternatively, use the heuristic detector (`src/heuristic_ack_detector.py`) for lightweight acknowledgment-only classification without an external LLM API.
3. **Evaluate pruning trade-offs.** Run `src/pruning_validation.py` with `--mode online_pruning` to measure the accuracy–efficiency trade-off of removing detected acknowledgment messages in live conversations. The statistical analysis script (`src/analysis.py`) produces McNemar tests, equivalence tests, and summary figures.

Y.3 License and Ethics

The code and annotation data are released under the MIT License. The underlying benchmark datasets (GSM8K, ARC-Challenge, StrategyQA) retain their original licenses. All three model backbones use permissive open-weight licenses. No personally identifiable information is contained in any released artifact; all data consists of LLM-generated text and researcher annotations thereof.

NeurIPS Paper Checklist

1. **Claims Yes**: All claims are supported by empirical evidence from our experiments. Ablation methodology is described as "perturbation-based" to accurately reflect its observational nature.

2. **Limitations Yes**: Section 6 discusses limitations including model scale (7B primary, substantive 14B oracle, 72B oracle N=265), topology coverage (two-agent with exploratory three-agent pilot), LLM annotation scope (κ variation), and greedy decoding sensitivity.
3. **Theory Assumptions and Proofs N/A**: Our work is empirical; no theoretical proofs are presented.
4. **Experimental Result Reproducibility Yes**: Section 4 specifies all models, hyperparameters, framework versions, and statistical tests. An anonymous code repository containing the complete experimental pipeline, configuration files, and sample data is provided at `[anonymized]`. GSM8K, ARC-Challenge, and StrategyQA are publicly available benchmark datasets.
5. **Open access to data and code Yes**: Code, traces, and annotations are provided in an anonymous repository at `[anonymized]` under the MIT License. GSM8K, ARC-Challenge, and StrategyQA are publicly available benchmark datasets.
6. **Experimental Setting/Details Yes**: Section 4 provides full experimental details including model names, parameter counts, generation settings, annotation protocol, and statistical framework.
7. **Experiment Statistical Significance Yes**: All experiments report exact p-values from McNemar exact tests, TOST equivalence tests, likelihood-ratio tests (LRT), and Fisher exact tests. The primary test family (9 LRT tests) receives Benjamini–Hochberg correction at $\alpha = 0.05$. Effect sizes, confidence intervals, minimum detectable effects, and statistical power are reported for all key comparisons. Ten-run online replication provides variance estimates and formal equivalence (Qwen $\Delta = -0.05 \pm 0.99$ pp, TOST $p = 0.007$ at ± 1 pp; 95% CI $[-0.75, +0.65]$ pp). Full statistical details appear in Appendix A.
8. **Compute Resources Yes**: Experiments were conducted on 4× NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each). Total compute including all replications, oracle validation, and ablations is approximately 120 GPU-hours.
9. **Code Of Ethics Yes**: This research studies existing open-weight models on a public benchmark. No new models are trained. Human annotation involved four research team members classifying LLM-generated messages; no external participants or sensitive data were involved.
10. **Broader Impacts Yes**: Our work characterizes communication in two-agent LLM systems to improve efficiency and interpretability. Our finding that structurally minimal messages (acknowledgments) are disproportionately impactful implies a potential adversarial vector: an attacker who can inject or suppress acknowledgment-like messages in a two-agent pipeline could cause outsized behavioral disruption with minimal detectable content. Defenses should monitor structural patterns, not just content, for integrity. Beyond this, we do not foresee negative societal impacts beyond those inherent to LLM deployment generally.
11. **Safeguards Yes**: The paper identifies acknowledgment injection as a potential adversarial vector (Section 6). This is a research finding, not a risk introduced by our methodology.
12. **Licenses Yes**: GSM8K is MIT-licensed. All three model backbones use permissive open-weight licenses. AG2/AutoGen is Apache 2.0.
13. **New Assets Yes**: We release interaction traces, function type annotations, and ablation results in the anonymous repository at `[anonymized]` under the MIT License. See Appendix Y for a complete artifact description.
14. **Crowdsourcing and Research with Human Subjects Yes**: Research team members classified 225 inter-agent messages into 5 communication function types for human validation of the LLM-derived taxonomy (Section 6). No external crowdsourcing was used. No personally identifiable or sensitive data were collected; annotators labeled only LLM-generated text.
15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects N/A**: The human annotation task involved only research team members classifying LLM-generated text, with no collection of personal data. This does not constitute human subjects research under standard IRB definitions.
16. **Declaration of LLM Usage Yes**: GPT-4o is used for cross-model annotation of communication function types (Section 4.1). Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct,

and Gemma-2-9B-IT serve as both backbone models generating conversations and cross-annotators. The annotation protocol, prompts, and inter-annotator agreement are documented in Section 4.1 and Appendices D, F. Human validation on 225 messages confirms taxonomy reliability ($\kappa = 0.76$). LLMs were not used for writing or editing the paper text.